# A380X flyPad failures

The A380X flyPad failure list says what each failure really fails, lists the landing gear sensor and actuator failures, and adds brake, trim air, feed tank pump, crossfeed valve and transfer pump failures, with the trim air duct overheat and fuel pump alerts of the FCOM, and the A380X engine failures: flameout and seizure with the in-flight relight envelope and the FCOM auto relight, ignition and start, oil, stall, EGT and vibration, FADEC, thrust lever, reverser and fuel filter clog failures, with their ECAM alerts and SD ENGINE page indications, the THR LEVERS NOT SET and OIL TEMP LO alerts, the thrust loss and FLEX TEMP not received FADEC failures, the E/WD N1 red limit with its red cross, the crossbleed engine start at X BLEED AUTO, the engine alert levels and procedures of the FCOM, braking and steering control failures, erroneous ADR airspeed and AOA failures with the PRIM air data monitoring, and the RCL / RCL LAST, T.O CONFIG NORMAL, automatic SD page and BTV INOP rules of the A380 FCOM

| | |
|---|---|
| **Aircraft** | A380X |
| **Where to find it** | flyPad > Failures (ATA 21, 24, 26, 28, 32, 34, 72, 73, 74, 77, 78, 79, 80); overhead FUEL, AIR and ENG MAN START panels; ENG MASTER levers and ENG START selector; thrust levers; brake pedals and tiller; ECAM control panel (RCL, RCL LAST, T.O CONFIG); E/WD, STATUS, PFD and the SD FUEL, COND, BLEED and ENGINE pages |
| **Branch** | `core/failures-a380x`, based on `master` (FlyByWire master 76db0aef1 merged in), with `core/fire` and `core/ecam-fws` merged in (2026-10-05, for the engine failures: they cut the fuel on the LP valve starvation of core/fire and complete the ENG FAIL ECAM procedures of core/ecam-fws). Merge core/fire and core/ecam-fws first, or with it. It also carries, as identical files: the shared engine failure modules of the A32NX engine failures (systems engine failure, start, oil, malfunction and engine control modules, the FADEC oil failures and the in-flight start throttle hold; the fbw-common part of `core/failures-a32nx`), the EWD EGT limits part of a `core/fcom-audit-fixes` fix, and the A380X crossbleed valve supply fix of `core/a380x-elec`. The ELEC EMER CONFIG alert itself is on `core/a380x-elec`: merged with it, that alert must read the FWS signal of this branch (inhibited by ALL ENG FLAME OUT). The fire and ECAM features are described in the core/fire and core/ecam-fws manuals. The RCL / RCL LAST part needs the EMER CANC pb and ECP MORE key of `core/ecam-fws` (merged here). Since 2026-10-12 it also carries, as identical files, the shared engine start and engine failure modules of the A32NX FCOM batch 7 and the shared overspeed fields of the A32NX LEAP-1A red line fix: the A380X sets the new values to its old behaviour, so they change nothing on the A380X. Two adaptations: the ENG TWO ENGS OUT deferred procedures read the FUEL JETTISON ARM / ACTIVE pbs directly (the jettison FWS is `core/fuel`), and T.O CONFIG NORMAL sees no ELEC GEN / DRIVE alert here (they are `core/a380x-elec`'s; both branches together are `core/a380x-elec-fuel-alerts`). |
| **Status** | Gear failures tested in the sim by the user: OK (2026-10-03). Brake / trim air failures and feed pump / crossfeed valve failures tested in the sim by the user: OK (2026-10-04); the trim air duct overheat / jammed valve alerts and the transfer pump failures with the fuel follow-ups too (2026-10-04). Clearer names: not sim-tested (labels only, checked by the a380.spec.ts unit tests). Engine failures: all tested on develop, the core branch itself was not sim-tested. Flameout / seizure / relight: OK 2026-10-05 and 2026-10-06 (two and four engines out). Auto relight, igniter indication, ground start with a failed igniter, stall / EGT / vibration, oil failures, FADEC network link and reverser unlocked: tested 2026-10-06; the three faults found then (restart after a shutdown, FADEC network link thrust, oil overheat) are fixed on this branch. HP fuel valve, restart after shutdown and reverse thrust: OK 2026-10-07. Core spike guard (E/WD N3 at the end of a relight), fuel filter clog, ENG MASTER FAULT light, SD start valve and the starter failure fix: OK 2026-10-08 (recorded). The other start, FADEC, thrust lever and reverser failures: not sim-tested (Rust, native and unit tests). Engine alerts 2026-10-08 (feature branch, recorded): THR LEVERS NOT SET, THRUST LOSS and T.O THRUST DISAGREE OK; OIL TEMP LO, the E/WD N1 and SD N3 red crosses, and the stall during a relight fix: |

| | |
|---|---|
| | not sim-tested; the N1 / FF / EGT part of the core spike guard: OK on develop 2026-10-09. 2026-10-12 picks, tested on the integration build: crossbleed start X1 / X2 and engine alert levels A1 / A2 OK (2026-10-09 evening), ADR air data A1 OK (2026-10-09/10 night flight), relight thrust pulse S1 and EGT OVER LIMIT with FADEC FAULT A3 OK (2026-10-10 morning); brake / steering B1 and RCL, T.O CONFIG NORMAL, SD pages N3 / N4 / N6 recorded as passed in the commit messages; BTV INOP not sim-tested. The core branch itself was not sim-tested. |
| **Pictures** | No pictures: the failures page, the ECAM alerts and the SD ENGINE page are the stock FBW displays. |

1. Landing gear sensor and actuator failures listed, ROLLOUT removed
2. Clearer failure names
3. Brake and trim air failures
4. Trim air duct overheat and jammed valve alerts
5. Feed tank pump and crossfeed valve failures
6. Transfer pump failures and fuel follow-ups
7. A380X engine failures (flameout, seizure, relight)
8. Automatic relight with both igniters, and the engine feed tank fuel draw
9. SD ENGINE page igniter indication
10. Engine ignition and start failures, FCOM ground start
11. HP fuel valve: no engine burning without a FADEC start
12. Engine oil failures (leak, filter clog, overheat)
13. Engine fuel filter clog, ENG MASTER FAULT light, SD start valve and N3 red cross
14. Engine stall, EGT overtemperature and vibration failures
15. FADEC, thrust lever and reverser failures
16. ENG THR LEVERS NOT SET and ENG OIL TEMP LO
17. FADEC thrust failures: ENG THRUST LOSS and ENG T.O THRUST DISAGREE
18. E/WD N1 red above 111 % with a red cross
19. Crossbleed engine start with X BLEED at AUTO
20. Engine alerts as in the FCOM
21. Braking and steering control failures
22. Erroneous ADR airspeed and AOA failures, PRIM air data monitoring
23. FCOM batch 7: RCL / RCL LAST, T.O CONFIG NORMAL, automatic SD pages, BTV INOP

## 1. Landing gear sensor and actuator failures listed, ROLLOUT removed

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 32 Landing Gear | **Status:** Tested in the sim by the user: OK (2026-10-03). Unit tests: a380.spec.ts.

### What it is for

The landing gear proximity sensor and jammed actuator failures (ids 32004 to 32025) were already simulated by the common FBW gear system and mapped in the A380X systems, but the flyPad listed only one of them, and with the wrong name: "uplock nose gear #1" failed the **right main gear** uplock sensor #1. All 18 failures are now listed, with the ids and names of the Rust failure map (the same as on the A32NX).

The **ROLLOUT** failure (ATA 22) only set an FCDC input that nothing reads (there is no steering control system yet). It is no longer listed; its id stays in the code so the C++ failure list keeps in step.

### What changed in the list

| Id | Before | Now |
|---|---|---|
| 32004 | not listed | Proximity sensor damage uplock nose gear #1 |
| 32005 | not listed | Proximity sensor damage downlock nose gear #2 |
| 32006 | Proximity sensor damage uplock nose gear #1 (but failed the right gear) | Proximity sensor damage uplock right gear #1 |
| 32007 | not listed | Proximity sensor damage downlock right gear #2 |
| 32008 | not listed | Proximity sensor damage uplock left gear #2 |
| 32009 | not listed | Proximity sensor damage downlock left gear #1 |
| 32010 to 32015 | not listed | Proximity sensor damage closed / opened nose, right, left gear door #1 / #2 (six entries) |

| | | |
|---|---|---|
| 32020 | not listed | Nose gear jammed actuator |
| 32021, 32022 | not listed | Left / Right main gears (body + wing) jammed actuator |
| 32023 | not listed | Nose gear door jammed actuator |
| 32024, 32025 | not listed | Left / Right main gear doors jammed actuator |
| 22001 | ROLLOUT (ATA 22) | not listed |

> **NOTE**
> A door *uplock* sensor ("closed ... door") detects the door closed, a door *downlock* sensor ("opened ... door") the door open. The 32021 / 32022 and 32024 / 32025 names come from the clearer names feature below: one failure jams the whole left or right main gear group.

### What you should see

| Situation | What you should see |
|---|---|
| **A jammed gear actuator failure set, gear lever moved** | That gear (or gear door) group stays where it was: the common FBW gear system freezes its actuator. |
| **A proximity sensor failure set** | That sensor gives its LGCIS a wrong signal. The gear logic and its indications are the existing FBW gear model: this branch only lists the failures. |
| **ATA 22 list** | No ROLLOUT entry. |

### How to use

1. Open flyPad > Failures > ATA 32 Landing Gear.
2. Pick the sensor or actuator failure by its name (the table above gives what each one fails) and activate it.
3. Deactivate it to clear the failure.

### How to test

| # | Do this | Expected result |
|---|---|---|
| 1 | Open flyPad > Failures > ATA 32 Landing Gear. | The 12 "Proximity sensor damage ..." and 6 "... jammed actuator" entries are listed, among them "Proximity sensor damage uplock right gear #1" and "Proximity sensor damage uplock nose gear #1". |
| 2 | Open flyPad > Failures > ATA 22 Auto Flight. | No ROLLOUT entry (FMC-A, FMC-B, FMC-C, FCU 1, FCU 2 are still there). |
| 3 | In flight, gear down, activate **Nose gear jammed actuator**, then set the gear lever UP. | The nose gear does not retract; the main gears do. |
| 4 | Deactivate the failure. | The nose gear retracts. |

## 2. Clearer failure names

> **Aircraft:** A380X | **Where:** flyPad > Failures | **Status:** Not sim-tested (labels only). Unit tests: a380.spec.ts.

## What it is for

The failure names were audited ATA by ATA against what the systems code fails (the Rust failure map, the C++ failure lists and the TypeScript consumers). Names that were misleading or misspelt now say what is failed. Only the labels change: the ids and the failures themselves are the same.

## Old name and new name

| ATA | Before | Now |
|---|---|---|
| 21 | Foward Cargo Isolation Valve / Foward Cargo Extraction Fan | Forward Cargo Isolation Valve / Forward Cargo Extraction Fan |
| 24 | TR ESS | ESS TR |
| 24 | TR APU | APU TR |
| 24 | AC 247XP | AC EHA (247XP) |
| 24 | DC 247PP | DC EHA (247PP) |
| 24 | DC 309PP | DC APU (309PP) |
| 26 | Engine 1 Loop A ... Engine 4 Loop B | Engine 1 fire detection loop A ... Engine 4 fire detection loop B |
| 26 | APU Loop A / APU Loop B | APU fire detection loop A / APU fire detection loop B |
| 26 | Main Landing Gear Bay Loop A / B | MLG bay fire detection loop A / B |
| 32 | LGCIU 1 Power supply, LGCIU 2 Power supply | LGCIS 1 power supply, LGCIS 2 power supply |
| 32 | LGCIU 1 Internal error, LGCIU 2 Internal error | LGCIS 1 internal error, LGCIS 2 internal error |
| 32 | Main left / right gear jammed actuator | Left / Right main gears (body + wing) jammed actuator |
| 32 | Main left / right gear door jammed actuator | Left / Right main gear doors jammed actuator |
| 34 | RA SYS A Interrupted (B, C) | RA SYS A antenna interrupted (B, C) |
| 34 | RA SYS A Direct Coupling (B, C) | RA SYS A antenna direct coupling (B, C) |
| 34 | TERR 1 / TERR 2 | TERR SYS 1 / TERR SYS 2 |

## Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **"AC ESS" (24105) and "AC EMER" (24104) keep their names.** | The A380X electrical model calls the real AC ESS busbar "AC ESS SHED" and the AC EMER busbar "AC ESS"; the labels already follow the real busbars. | Use "AC ESS" for the AC ESS busbar, "AC EMER" for the AC EMER busbar. |
| **"XPDR 2" (34004) and "NSS AVNCS ANSU 2" (46002) keep their names but fail nothing on this branch.** | Nothing reads them on master. They are wired on `core/a380x-elec`. | Test them with that branch. |

## How to use

1. Open flyPad > Failures and pick the failure by its new name (table above).

## How to test

| # | Do this | Expected result |
|---|---|---|

| | | |
|---|---|---|
| 1 | Open flyPad > Failures > ATA 24 Electrical Power. | The entries read "ESS TR", "APU TR", "AC EHA (247XP)", "DC EHA (247PP)", "DC APU (309PP)". |
| 2 | Open ATA 26 Fire Protection. | The loop failures read "Engine 1 fire detection loop A" ... "MLG bay fire detection loop B". |
| 3 | Open ATA 32 Landing Gear and ATA 34 Navigation. | "LGCIS 1 power supply" ... "LGCIS 2 internal error"; "RA SYS A antenna interrupted", "RA SYS A antenna direct coupling", "TERR SYS 1", "TERR SYS 2". |
| 4 | Open ATA 21 Air Conditioning. | "Forward Cargo Isolation Valve" and "Forward Cargo Extraction Fan" are spelt right. |
| 5 | Activate "ESS TR" with the aircraft powered. | The ESS TR fails, as "TR ESS" did before (same id 24002). |

## 3. Brake and trim air failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 32 and ATA 21 | **Status:** Tested in the sim by the user: OK (2026-10-04); the brake leak works but shows nothing on the SD HYD page.

### What it is for

The A380X hydraulic and air conditioning models already had brake leak and trim air faults, but the flyPad had no failure for them. Eleven failures are now listed (same ids as on the A32NX for the brakes), and a spec checks that every id of the systems failure map is listed with its label.

| flyPad name (ATA) | What it fails |
|---|---|
| Brakes NORM circuit leak (green hydraulic) (32) | The NORM brakes are fed by the GREEN system in this model: GREEN fluid leaks out (0.1 gal/s) while the circuit is pressurised. |
| Brakes ALTN circuit leak (yellow hydraulic) (32) | The ALTN brakes (with the brake accumulator) are fed by the YELLOW system: YELLOW fluid leaks out while pressurised. |
| Brakes ALTN accumulator gas leak (32) | The ALTN brake accumulator loses its gas charge: the stored brake pressure drops when the YELLOW system is not pressurised. |
| Cockpit / Main deck zone 1 / Upper deck zone 1 / Forward cargo trim air valve jammed (21) | The trim air valve of that zone stays where it was when the failure was set. |
| Cockpit / Main deck zone 1 / Upper deck zone 1 / Forward cargo duct overheat (hot trim air) (21) | 200 deg C air is blown into the zone duct while its hot air valve is open. The zone gets too hot and the controller closes the trim air valves; HOT AIR OFF stops the overheat (A380 FCOM COND DUCT OVHT procedure). |

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **No ECAM alert and no SD indication for the brake failures.** | The SD HYD page has no brake leak indication. The trim air failures now have their alerts and SD marks: see "Trim air duct overheat and jammed valve alerts" below. | Watch the hydraulic quantity / pressure and the brake pressure. |
| **A full MSFS restart is needed after an update.** | The systems WASM module changes. | Restart the sim after installing. |

### How to use

1. Brakes: flyPad > Failures > ATA 32 Landing Gear, the three "Brakes ..." entries.

2. Trim air: flyPad > Failures > ATA 21 Air Conditioning, the "... trim air valve jammed" and "... duct overheat (hot trim air)" entries.

### How to test

| # | Do this | Expected result |
|---|---|---|
| 1 | A380X on the ground, four engines running (GREEN and YELLOW pressurised). Activate **Brakes NORM circuit leak (green hydraulic)**. | The GREEN system loses fluid; YELLOW is not affected. Nothing is shown for the leak itself on the SD HYD page. |
| 2 | Deactivate it, activate **Brakes ALTN circuit leak (yellow hydraulic)**. | The YELLOW system loses fluid; GREEN is not affected. |
| 3 | Set the parking brake, then stop the engines (YELLOW no longer pressurised). Activate **Brakes ALTN accumulator gas leak**. | The brake pressure held by the accumulator drops. |
| 4 | In flight, HOT AIR pushbuttons ON, activate **Cockpit duct overheat (hot trim air)**. Open the SD COND page. | The cockpit zone gets hotter; then the trim air valves close. Setting HOT AIR OFF stops the overheat. |
| 5 | Activate **Cockpit trim air valve jammed**, then change the cockpit temperature selection. | The cockpit trim air valve no longer moves: the cockpit temperature stops following the selection; the other zones still regulate. |

## 4. Trim air duct overheat and jammed valve alerts

> **Aircraft:** A380X | **Where:** overhead AIR panel (HOT AIR 1 / 2, PACK 1 / 2); E/WD, STATUS; SD COND page | **Status:** Tested in the sim by the user: OK (2026-10-04). Rust tests (air_conditioning), unit tests: CondTrimAirFlags.spec.ts, CondTrimAirMarks.spec.ts.

### What it is for

The trim air failures above now give the alerts of the A380 FCOM (PRO-ABN-ECAM-10-21-10). The temperature controller monitors each trim air duct and valve: a duct above 70 deg C for 1.2 s is an overheat, latched until its HOT AIR pushbutton is set OFF; a valve that does not follow its demand is jammed. The FWS reads both from the controller and raises:

- **COND DUCT OVHT** (master caution, single chime, FAULT light in the HOT AIR pushbutton, SD COND page) with the CKPT DUCT OVHT, CABIN DUCT OVHT and FWD CARGO DUCT OVHT cases, the deferred procedure **WHEN DUCT OVHT OUT**, and its STATUS lines (INOP SYS, CABIN TEMP REGUL DEGRADED, FWD CRG TEMP REGUL DEGRADED).
- **COND FWD CARGO TEMP REGUL FAULT** for a jammed forward cargo trim air valve: CARGO TRIM AIR VLV FAULT, CARGO TEMP ... MONITOR, STATUS INFO FWD CRG TEMP REGUL DEGRADED (SD COND page only: no chime, no master light).
- SD COND page: the amber OVHT mark of an overheated cockpit or cabin zone duct and the trim air valve mark of a jammed valve (before: never shown).

### What you should see

| Situation | What you should see |
|---|---|
| Cockpit duct overheat | COND DUCT OVHT, CKPT DUCT OVHT: HOT AIR 2 ... OFF; if the hot air valve stays open (HOT AIR JAMMED OPEN): PACK 2 ... OFF. HOT AIR 2 FAULT light, OVHT on the SD COND page. |
| Main deck zone duct overheat | CABIN DUCT OVHT: HOT AIR 1 ... OFF (HOT AIR 2 for an upper deck zone); if jammed open, PACK 1 (2) ... OFF; CARGO TEMP ... MONITOR. |

| | |
|---|---|
| **Forward cargo duct overheat** | FWD CARGO DUCT OVHT: HOT AIR 1 ... OFF; if jammed open, PACK 1 ... OFF. |
| **Overheat gone (duct cooled, HOT AIR OFF)** | Deferred procedure WHEN DUCT OVHT OUT: HOT AIR 1 (2) ... ON, and PACK 1 (2) ... ON if it was switched off. |
| **Forward cargo trim air valve jammed** | COND FWD CARGO TEMP REGUL FAULT; STATUS INFO FWD CRG TEMP REGUL DEGRADED; the valve mark on the SD COND page. |

## Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **Which HOT AIR valve supplies which zone is a design choice.** | The FCOM does not say it: the main deck zones and the forward cargo are on HOT AIR 1, the cockpit and the upper deck on HOT AIR 2. | - |
| **A jammed forward cargo valve counts as a degraded regulation (no INOP SYS).** | Design choice: the aircraft has two forward cargo trim air valves and the FCOM says a jammed valve loses the "optimized" regulation. | - |
| **The alerts are inhibited during the takeoff and the landing (FWS phases 3 to 7, 9, 10).** | The FCOM gives the inhibition as a figure: that of COND HOT AIR FAULT / FWD CARGO ISOL FAULT is used. | Test on the ground before takeoff or in cruise. |
| **A full MSFS restart is needed after an update.** | The systems WASM module changes. | Restart the sim after installing. |

## How to use

1. Open flyPad > Failures > ATA 21 Air Conditioning and activate a "... duct overheat (hot trim air)" or a "... trim air valve jammed" failure.
2. Follow the ECAM with the HOT AIR and PACK pushbuttons of the overhead AIR panel; check the SD COND page.
3. Deactivate the failure; when the duct has cooled, follow WHEN DUCT OVHT OUT.

## How to test

| # | Do this | Expected result |
|---|---|---|
| 1 | A380X in cruise (or on the ground before takeoff), packs and HOT AIR 1 / 2 ON. Activate **Cockpit duct overheat (hot trim air)**. | After the duct passes 70 deg C: master caution, E/WD COND DUCT OVHT, CKPT DUCT OVHT with HOT AIR 2 ... OFF; HOT AIR 2 FAULT light; SD COND page with the amber OVHT mark on the cockpit. |
| 2 | Set HOT AIR 2 OFF. Open the STATUS page. | The line is done; STATUS INOP SYS and INFO lines of COND DUCT OVHT. |
| 3 | Deactivate the failure and wait for the duct to cool. | The alert goes; WHEN DUCT OVHT OUT asks HOT AIR 2 ... ON. |
| 4 | Activate **Main deck zone 1 duct overheat (hot trim air)**. | COND DUCT OVHT, CABIN DUCT OVHT with HOT AIR 1 ... OFF and CARGO TEMP ... MONITOR. |
| 5 | Deactivate it. Activate **Forward cargo trim air valve jammed**, then change the cargo temperature selection. | COND FWD CARGO TEMP REGUL FAULT (no chime): CARGO TRIM AIR VLV FAULT, CARGO TEMP ... MONITOR; STATUS INFO FWD CRG TEMP REGUL DEGRADED. |
| 6 | Activate **Cockpit trim air valve jammed** and change the cockpit temperature selection. | The trim air valve mark appears on the cockpit zone of the SD COND page. |

## 5. Feed tank pump and crossfeed valve failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; SD FUEL | **Status:** Tested in the sim by the user: OK (2026-10-04), after the fix that holds the MSFS pump / valve switch.

### What it is for

Twelve new failures: the main and standby pump of each feed tank and the four crossfeed valves. The FUEL panel pushbuttons now set a selection, and the systems host moves the MSFS pump or valve switch from it: a failed pump is held off, a jammed valve is held where it was. The FAULT lights, the SD FUEL page and the FCOM alerts compare the selection with the real state (A380 FCOM PRO-ABN-ECAM-10-28, DSC-28-20).

| flyPad name (ATA 28) | What it fails |
|---|---|
| Feed tank 1 main pump ... Feed tank 4 main pump | That main pump stops (held off). |
| Feed tank 1 standby pump ... Feed tank 4 standby pump | That standby pump stops (held off). |
| Crossfeed valve 1 jammed in place ... Crossfeed valve 4 jammed in place | That crossfeed valve stays where it was. |

### What you should see

| Situation | What you should see |
|---|---|
| Feed pump failed, its pushbutton ON | FAULT light in the pushbutton; the pump amber on the SD FUEL page. |
| One main pump failed | **FUEL FEED TK 1 MAIN PMP FAULT** (tank 1 to 4), SD FUEL page; procedure FEED TK 1 MAIN PMP ... OFF; STATUS INOP SYS FEED TK 1 MAIN PMP. |
| One standby pump failed | **FUEL FEED TK 1 STBY PMP FAULT**, crew awareness; STATUS MORE: REDUND LOSS FEED TK 1 STBY PMP. |
| Main and standby pumps of one tank failed | **FUEL FEED TK 1 MAIN + STBY PMPs FAULT** (replaces the two single alerts). After takeoff: CROSSFEED 1 ON, CROSSFEED 2 ON (CROSSFEED 3 and 4 for tanks 3 and 4), FEED TK 1 MAIN PMP OFF, FEED TK 1 STBY PMP OFF, AVOID NEG G LOAD; before takeoff only the two pump lines. STATUS: INOP SYS FEED TK 1 PMPs, FEED TK 1 BY GRVTY ONLY. |
| Crossfeed valve jammed and its pushbutton moved | The valve disagrees with its pushbutton: FAULT light, valve amber on the SD FUEL page, **FUEL CROSSFEED VLV 1 FAULT** (crew awareness), STATUS INOP SYS CROSSFEED VLV 1; if the valve is stuck closed and engine 1 is not running, FEED TK 1 NOT USABLE. |

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **The alerts are inhibited during the takeoff and the landing.** | The FCOM gives the inhibition as a figure: the one of JETTISON VLV NOT CLOSED is used (FWS flight phases 4 to 7, 9 and 10). | Test on the ground before takeoff or in cruise. |
| **A pump turned OFF by the crew also gives the FAULT alert.** | FCOM: a pump is faulty when it runs at low pressure, is abnormally not running, or is turned off by the crew (except in flight phases 1 and 12). | Expected behaviour. |
| **The gravity feed ceiling (MAX FL 280, 50/MEA) is not simulated.** | No gravity feed model. | None. |
| **A full MSFS restart is needed after an update.** | The cockpit behaviour XML changes. | Restart the sim after installing. |

### How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate a pump or crossfeed valve failure.
2. Follow the ECAM procedure with the overhead FUEL panel pushbuttons; check the SD FUEL page.
3. Deactivate the failure to restore the pump or valve.

### How to test

| # | Do this | Expected result |
|---|---|---|
| 1 | A380X on the ground, engines running, all feed pumps ON, crossfeed valves in their normal position. Open the SD FUEL page. | No FAULT light, no fuel alert. |
| 2 | Activate **Feed tank 1 main pump**. | FAULT light in the FEED TK 1 MAIN pushbutton, pump amber on the SD, E/WD: FUEL FEED TK 1 MAIN PMP FAULT with FEED TK 1 MAIN PMP ... OFF. |
| 3 | Set the FEED TK 1 MAIN pushbutton OFF. Open the STATUS page. | The procedure line is done; STATUS INOP SYS: FEED TK 1 MAIN PMP. |
| 4 | Set it ON again and also activate **Feed tank 1 standby pump**. | The single alerts are replaced by FUEL FEED TK 1 MAIN + STBY PMPs FAULT (on the ground: the two pump OFF lines); STATUS: FEED TK 1 PMPs, FEED TK 1 BY GRVTY ONLY. |
| 5 | Deactivate both. Activate **Feed tank 2 standby pump** only. | FUEL FEED TK 2 STBY PMP FAULT; STATUS MORE: REDUND LOSS FEED TK 2 STBY PMP. |
| 6 | Deactivate it. Activate **Crossfeed valve 1 jammed in place**, then move the CROSSFEED 1 pushbutton. | The valve does not move: FAULT light, valve amber on the SD, FUEL CROSSFEED VLV 1 FAULT, STATUS INOP SYS CROSSFEED VLV 1. |
| 7 | Deactivate the failure. | The valve follows its pushbutton again; the alert clears. |

## 6. Transfer pump failures and fuel follow-ups

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28 Fuel; overhead FUEL panel; E/WD, STATUS; SD FUEL page | **Status:** Tested in the sim by the user: OK (2026-10-04). Unit tests: FuelPumpValveFlags.spec.ts, FuelPumpsAndValves.spec.ts, failable-fuel-switches.spec.ts.

### What it is for

Eight new failures of the transfer pumps that the FBW fuel transfers use: **L outer tank pump**, **R outer tank pump**, **L mid tank fwd pump**, **R mid tank fwd pump**, **L inner tank fwd pump**, **R inner tank fwd pump**, **Trim tank L pump**, **Trim tank R pump**. A failed pump is held off whatever the fuel system commands. The pushbutton FAULT light, the amber pump on the SD FUEL page and the FUEL ... PMP FAULT alerts of the A380 FCOM (PRO-ABN-ECAM-10-28) come as soon as a pump fails, runs in an empty tank with its pushbutton ON, or is switched OFF by the crew (except in flight phases 1 and 12). A repaired pump follows the automatic transfers again.

- **FUEL FEED TK 1(2)(3)(4) LEVEL LO** now triggers for a single low feed tank: the four tanks shared one 30 s confirmation, so only all four low together could trigger it.
- The **CROSSFEED** pushbutton OPEN light now follows its valve.
- The sensed CROSSFEED ... ON and ALL FEED TKs PMPs ON items of the ECAM procedures tick on the pushbuttons, even with a jammed valve or a failed pump.
- Corrupted degree signs in the ECAM texts are fixed.

## What you should see

| Situation | What you should see |
|---|---|
| L outer tank pump failed | FAULT light; **FUEL L OUTR TK PMP FAULT**: L OUTR TK PMP ... OFF; STATUS: OUTR TKs XFR BY GRVTY ONLY, INOP SYS L OUTR TK PMP. |
| L inner (mid) tank fwd pump failed | **FUEL L INR (MID) TK FWD PMP FAULT**: L INR (MID) TK FWD PMP ... OFF; STATUS INOP SYS L INR (MID) TK FWD PMP. |
| Its aft pump switched OFF too | **FUEL L INR (MID) TK FWD+AFT PMPs FAULT** replaces the single alerts: both pumps ... OFF; L INR (MID) TK NOT USABLE; INOP SYS L INR (MID) TK PMPs. |
| Trim tank L and R pumps failed | **FUEL TRIM TK L+R PMPs FAULT**: both pumps ... OFF, PITCH ATT < 3° FOR GRVTY FWD XFR, TRIM TK FEED ... OPEN, TRIM TK : 2300 KG MAX NOT USABLE; STATUS: TRIM TK XFR BY GRVTY ONLY, TRIM TK GRVTY XFR : AROUND 3000 KG / HR, INOP SYS TRIM TK PMPs. |
| One feed tank low | FUEL FEED TK n LEVEL LO for that tank after 30 s. |
| CROSSFEED pushbutton ON | The OPEN light comes on when the valve is open. |

## Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **The aft gallery pumps (inner and mid tank aft) cannot fail.** | They never run in the simulation; switching them OFF still gives their alert. | - |
| **No gravity transfer.** | Not simulated: the STATUS gravity lines are information only. | None. |
| **The alerts are inhibited during the takeoff and the landing.** | Same inhibition as the feed pump alerts above. | Test on the ground before takeoff or in cruise. |
| **A full MSFS restart is needed after an update.** | The cockpit behaviour XML changes (OPEN light). | Restart the sim after installing. |

## How to use

1. Open flyPad > Failures > ATA 28 Fuel and activate one of the eight transfer pump failures.
2. Follow the ECAM with the overhead FUEL panel; check the SD FUEL page.
3. Deactivate the failure to restore the pump; it follows the automatic transfers again.

## How to test

| # | Do this | Expected result |
|---|---|---|
| 1 | A380X on the ground before takeoff (or in cruise) with fuel in the outer tanks, all fuel pumps ON. Activate **L outer tank pump**. | FAULT light in the L OUTR TK pushbutton, pump amber on the SD FUEL page, master caution, E/WD FUEL L OUTR TK PMP FAULT with L OUTR TK PMP ... OFF. |
| 2 | Set the L OUTR TK pushbutton OFF. Open STATUS. | The line is done; STATUS: OUTR TKs XFR BY GRVTY ONLY, INOP SYS L OUTR TK PMP. |
| 3 | Deactivate it and set the pushbutton ON. Activate **L inner tank fwd pump**, then set the L INR TK AFT pushbutton OFF. | FUEL L INR TK FWD+AFT PMPs FAULT replaces the single alert; L INR TK NOT USABLE. |
| 4 | Restore everything. Activate **Trim tank L pump** and **Trim tank R pump**. | FUEL TRIM TK L+R PMPs FAULT with its procedure; the degree sign of PITCH ATT < 3° is shown correctly. |
| 5 | Restore everything. Set a CROSSFEED pushbutton ON. | Its OPEN light comes on with the valve. |

## 7. A380X engine failures (flameout, seizure, relight)

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 72 Engine; pedestal ENG MASTER 1-4 and ENG START selector; overhead AIR panel X BLEED selector; E/WD, STATUS, SD ENGINE page | **Status:** Sim-tested OK on develop, recorded: 2026-10-05 ground card T1-T3 and in-flight card T1-T3 (windmilling, quick and crossbleed relights without APU), earlier T1, T7, T8 (relight with APU bleed) and T9; 2026-10-06 relight batch (quick relight of engine 4 levers in CL: stops at idle, then accelerates), Flight A S2 T2 (starter duct vents after the start), S4 T2 (two engines out, windmilling relight at 255 kt) and S6 T1 (all engines out: ALL ENG FLAME OUT, RAT, no ELEC EMER CONFIG on develop, relight). The core spike guard (the MSFS core jump at the end of a relight, below): sim-tested OK on develop 2026-10-08 (E/WD N3 held at 61.6 % while MSFS jumped to 96.8 %); its N1 / FF / EGT hold: OK on develop 2026-10-09 (FL150 relight: E/WD N1 held at 26.4 % and FF at 615 while MSFS N1 went to 62.5 %). The softened thrust pulse (MSFS core limiter, 2026-10-12 pick): tested in the sim by the user on the integration build, OK (2026-10-10 morning, batch 2 test S1: engine 4 relight at FL150), native test msfs_core_limiter_test. The stall during a relight fix: not sim-tested (installed 2026-10-08, no visible effect expected; native test R12). Rust tests (a380_systems engine_failure.rs, engine_out_aircraft_tests.rs, pneumatic.rs, hydraulic), native tests (relight_start_test, engine_start_throttle_hold_test), unit tests EngineFailAlerts.spec.ts, FwsAbnormalSensedItems.spec.ts, a380.spec.ts. The core branch itself was not sim-tested.

### What it is for

The A380X port of the A32NX engine failures. Eight new failures in the flyPad chapter ATA 72 Engine. A **flameout** cuts the engine fuel once: the crew relights the engine with ENG MASTER OFF then ON inside the in-flight relight envelope of the A380 FCOM. A **seizure** also stops the core: no relight while the failure is active. The FWS gives the FCOM ENG FAIL, ENG SHUT DOWN, TWO ENGS OUT and ALL ENG FLAME OUT alerts for the four engines.

| Id | flyPad failure (ATA 72 Engine) | What the code does |
|---|---|---|
| 72000-72003 | Engine 1 (2)(3)(4) flameout (crew relight possible) | Cuts the engine fuel (latched, MSFS fuel valve 60-63). The engine runs down and windmills. ENG MASTER OFF then ON inside the envelope relights it; it then runs normally while the failure stays listed. Deactivate and activate it again for a new flameout. |
| 72010-72013 | Engine 1 (2)(3)(4) seizure (no relight) | Cuts the fuel and stops the core (N3). No relight while the failure is active; once it is cleared the engine relights by the normal procedure. |

### What changed after the first sim tests

- **No N3 overshoot at the end of an in-flight start.** The three in-flight relights of 2026-10-05 took N3 from 61 % to 97.5 % for 5 to 7 s with the levers in CL: while the engine was out the FADEC thrust loop wound the MSFS throttle up to 100 %. Now the MSFS throttle stays at idle while the engine is out or starting and until it runs at idle, then the engine accelerates to the lever or A/THR demand (A380 FCOM DSC-70-30 "FADEC stabilizes the engine at idle speed"; ENG RELIGHT IN FLIGHT: THR LEVER IDLE, then WHEN IDLE REACHED).
- **A compressor stall no longer ends that idle hold early.** The hold ends when the commanded N1 reaches idle; with the "Engine n compressor stall" failure active at the end of a relight, the stall N1 loss (up to 15 %) was counted in the commanded N1 and the hold ended about 0.8 s early. The stall N1 loss is now taken out of that check (native test R12: FL200, 300 kt relight, levers in CL, with a stall).
- **TWO ENGS OUT STATUS** lists the FCOM lines PART SPLRs, PART L/G RETRACTION, G (Y) HYD SYS, PACK 1 (2) and one GEN line for the two generators lost (GEN 1+2, 3+4 on the same side; GEN 1+3, 1+4, 2+3, 2+4 on opposite sides).
- **ALL ENG FLAME OUT**: its RAT MAN ON line is sensed (it turns green when the RAT is deployed), it has the missing L/G GRVTY (EXTN 2 MIN) DOWN line before a forced landing, and it inhibits ELEC EMER CONFIG as in the FCOM. The ELEC EMER CONFIG alert itself is on `core/a380x-elec`: this branch only computes the inhibited signal, so the inhibition shows once both branches are merged.
- **ENG RELIGHT IN FLIGHT** shows the single engine limits MAX GUARANTEED ALTITUDE : 30000 FT and MIN SPEED FOR WINDML RELIGHT : 260 KT (FCOM I.174902-174904).

- **No starter air without a source.** A starter assisted relight was granted with no APU or crossbleed air: the air trapped in a stopped engine bleed duct and the slowly venting starter duct (about 45 s) kept the starter pressurised. The duct of an engine with no air source now empties within about 3 s, and the starter duct is empty (below 5 PSIG) about 3 s after the start valve closes (design values: the FCOM gives none).
- **X BLEED selector OPEN opens the crossbleed valves.** Their manual control used the A320 DC ESS SHED busbar, which the A380 does not have; it now uses DC ESS (A380 FCOM DSC-36-30). The crossbleed starter air of a starter assisted relight depends on it.
- **The FADECs read the ENG START selector.** They read a non-existent variable (always 0) and never saw the selector at IGN START; a static check now guards the variable names.
- **Each FADEC reads the LGCIU of its side**: engines 1 and 2 LGCIU 1, engines 3 and 4 LGCIU 2 (design choice, the FCOM does not say). Before, the on-ground input of engines 3 and 4 was read past the end of the two-LGCIU list.
- **ENG MASTER OFF closes the HP fuel valve** and the engine stops burning at once: see "HP fuel valve" below.
- **The thrust pulse at the end of a relight is softened** (2026-10-12). At the end of an in-flight start MSFS jumps its core to its top speed for a few seconds (recorded 2026-10-08: 57.7 to 96.2 % in 0.5 s, thrust from -2 100 to +11 800 lb on the relit engine, a yaw kick). The core spike guard only hid it from the indications. A new FADEC limiter now holds the MSFS core near idle until it has settled, so the real thrust pulse is gone.

### FCOM references

| A380 FCOM (a380_fcom.txt) | What it says | Used for |
|---|---|---|
| PRO-ABN-ECAM-10-70 ENG RELIGHT IN FLIGHT, l.174864-175027, chart PDF p.5854 | Single engine: "MAX GUARANTEED ALTITUDE : 30000 FT", "MIN SPEED FOR WINDML RELIGHT : 260 KT"; multiple engines: 28 000 ft and 250 kt; ENG START SEL IGN START; "Engine must relight within 30 s after fuel flow increases". | The relight envelope and the 30 s attempt |
| DSC-70-30 IN FLIGHT, l.113552-113575 | Engine start valve opens "if N2 is below 11 %, or if the aircraft airspeed (CAS) is below 260 kt"; it closes above 58.4 %. | The start valve in flight and for a restart |
| DSC-70-80-30-20, l.112539-112571 | Quick relight: MASTER ON-OFF-ON within 30 s, "provided that N2 is greater than 45 %"; starter assisted below 11 % N2, windmilling at or above 260 kt; during a windmilling start the FADEC disconnects both hydraulic pumps. | Quick relight, start type, engine pumps |
| ENG 1(2)(3)(4) FAIL, l.171712-171981 | Triggered when the engine is not running; "If more than two engines failed: ENG (AFFECTED) RELIGHT PROC ... APPLY" (l.171871). | ENG FAIL and RELIGHT PROC APPLY |
| ENG 1(2)(3)(4) SHUT DOWN, l.172693-172806 | Shown when the ENG MASTER is OFF or the ENG FIRE pb pushed; STATUS INOP SYS BTV (engine 2 or 3), CAT 3 DUAL and CAT 3 SINGLE ONLY if the APU is off. | ENG SHUT DOWN STATUS |
| ENG TWO ENGS OUT ON SAME / OPPOSITE SIDE, l.175216-175635 | Two engines failed on the same side / on opposite sides, LAND ANSA, STATUS (CAT 2 / CAT 3 INOP, CAT 2 ONLY). | The two cautions |
| ENG ALL ENG FLAME OUT, l.114538 | The title of the all-engines procedure. | Title of the old ALL ENGINES FAILURE |

The relight envelope, read off the chart (IAS, flight level):

| Zone | Where | Needs |
|---|---|---|
| 3, windmilling start | 260-340 kt up to FL300 (multiple engines out: from 250 kt, up to FL280) | ENG START IGN START, ENG MASTER OFF then ON; the start valve stays closed |

| | | |
|---|---|---|
| 1 and 2, starter assisted relight | 150-260 kt up to FL200; 185-260 kt from FL200 to FL250 | ENG START IGN START, ENG MASTER OFF then ON, and starter air: APU BLEED, or X BLEED OPEN with the other engines |
| 2 (and 3), quick relight | 220-260 kt up to FL200 (and zone 3), N3 above 45 % | ENG MASTER OFF then ON within 30 s, at any ENG START position |

## What you should see

| Situation | What you should see |
|---|---|
| **Flameout in flight** | N3, EGT and FF run down, FF 0, N1 / N3 settle at the windmill values. Amber **ENG n FAIL** with its FCOM procedure lines (THR LEVER n IDLE reads its own lever; ENG n RELIGHT PROC CONSIDER). |
| **ENG MASTER OFF in flight** | ENG n FAIL goes, **ENG n SHUT DOWN** comes with its STATUS: INOP SYS BTV for engine 2 or 3, CAT 3 DUAL and CAT 3 SINGLE ONLY with the APU off; the engine 4 bleed INOP comes from engine 4 (it came from another engine before). |
| **Windmilling relight (zone 3, no APU)** | At the MASTER ON the fuel comes back; engine at idle in about 35 s; the start valve stays closed; both engine pumps of that engine are off-loaded during the start. |
| **Crossbleed relight (zone 1, X BLEED OPEN)** | The start valve opens and the starter is pressurised in about 0.5 s; engine at idle in about 35 s; the start valve closes at N3 about 58 %. |
| **Quick relight (ENG START at NORM)** | MASTER OFF then ON at once: the engine relights from about 75 % N3, at idle in about 25 s. |
| **Any in-flight relight with the levers in CL (A/THR ON)** | N3 reaches idle (about 62 %) and holds there until the engine runs, then rises in about 5 s to the N3 of the other engines; it does not go more than 1-2 % above them. |
| **ENG RELIGHT IN FLIGHT procedure** | Single engine: MAX GUARANTEED ALTITUDE : 30000 FT, MIN SPEED FOR WINDML RELIGHT : 260 KT; multiple engines: 28000 FT and 250 KT. |
| **Two engines failed** | ENG TWO ENGS OUT ON SAME SIDE or ON OPPOSITE SIDE caution with LAND ANSA; the multiple-engine envelope applies. STATUS: PART SPLRs, PART L/G RETRACTION, G (Y) HYD SYS, PACK 1 (2), the GEN line of the two generators lost, CAT 2 / CAT 3 INOP, CAT 2 ONLY. |
| **Three engines failed** | ENG FAIL shows RELIGHT PROC APPLY instead of CONSIDER. |
| **All four engines failed** | Red ENG ALL ENG FLAME OUT (the FCOM title; it was ALL ENGINES FAILURE). RAT MAN ON turns green once the RAT is out; L/G GRVTY (EXTN 2 MIN) DOWN before a forced landing. With `core/a380x-elec` merged: no ELEC EMER CONFIG while it is shown. |
| **Starter assisted relight below 260 kt without APU BLEED and with X BLEED at AUTO** | No starter air: the engine does not relight (the starter duct is empty). Set X BLEED OPEN. |
| **On the ground** | A failed engine starts again as a normal start, with APU air (ENG START IGN START, MASTER OFF then ON). Two engines shut down for taxi give no TWO ENGS OUT caution. |

## Design choices

- The envelope governs **every** in-flight relight, also after a crew shutdown without any failure: MSFS would otherwise relight anywhere.
- The quick relight also works in zone 3, where the windmill alone lights the engine.

- A windmilling engine keeps N3 = 11 % x IAS / 260 kt and N1 = 1.5 x N3 (the FCOM gives no values).
- FBW models the Trent 972 (three spools): the FCOM N2 (GP7200 core) is read on the Trent N3. ENG FAIL keeps the N3 < 50 % threshold of FBW (the FCOM says N2 below 61 %).
- TWO ENGS OUT is shown only with exactly two engines out (with three, ENG FAIL RELIGHT PROC APPLY; with four, ALL ENG FLAME OUT).
- During a windmilling start the two engine pumps are off-loaded (depressurised), not disconnected: the ENG PUMPS DISC clutch cannot be reconnected in flight.
- Once a relight is granted in flight the FADEC holds the MSFS core at 25 % N2: MSFS only burns above 20 % N2, and its starter only turns with the APU bleed, so without it a windmilling or crossbleed relight would never burn.
- The throttle hold of an in-flight start (idle until the engine runs) works in flight only; a fallback ends it for a start that settles just below the FBW idle N3.
- With the ENG MASTER OFF in flight the MSFS core runs down to the windmill speed: also so on develop, not a fault.

## Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **ENG TWO ENGS OUT ON OPPOSITE SIDE has no PACK line in its procedure.** | Not added yet (FCOM I.175519); its STATUS has the PACK 1 (2) INOP SYS line. | - |
| **The MSFS core jump at the end of a relight is held, not removed.** | At the end of a start the MSFS engine model jumps its core from its low idle (60 % corrected) to the top of its fuel flow table (about 97 %) for 2 to 6 s. The FADEC filters this jump for the indications (core spike guard, 2026-10-08): the E/WD N1/N3, fuel flow, EGT and the FADEC logic stay at idle (sim-tested: MSFS 96.8 %, E/WD 61.6 %); the guard also holds the N1 (and so the fuel flow and EGT) while the MSFS fan follows the core jump a few frames late. Since 2026-10-12 the MSFS core limiter also holds the MSFS core near idle, so the thrust pulse of about 10 s recorded on 2026-10-08 (yaw rate at most 0.2 deg/s) is gone (sim test S1 2026-10-10: passed). A ground start shows no jump (recorded 2026-10-08). | None. |
| **Multiple engines out: the windmilling relight needs at least 250 kt.** | FCOM envelope (sim-confirmed 2026-10-06). | Descend to keep the speed. |
| **A full MSFS restart is needed after an update.** | The systems, FADEC and fbw WASM modules change. | Restart the sim after installing. |

## How to use

1. Open flyPad > Failures > ATA 72 Engine and activate a flameout or a seizure.
2. Follow the ECAM. To relight: get inside the envelope (table above), ENG START selector (pedestal, engine panel) IGN START, ENG MASTER OFF for 30 s, then ON. Below 260 kt without the APU, set the X BLEED selector (overhead AIR panel) to OPEN for starter air.
3. For a quick relight, set the ENG MASTER OFF then ON at once (within 30 s, N3 above 45 %), at any ENG START position.
4. When the engine is at idle: ENG START NORM, X BLEED AUTO; clear the failure in the flyPad (a seizure must be cleared before a relight).

## How to test

Short card in the sim-test format (from `reports/test-card-b1-inflight.md`, `test-card-b1-ground-2026-10-05.md`, `test-card-relight-batch-2026-10-06.md` and the S4 / S6 sessions of `test-plan-integration-2026-10-06.md`). Needs one MSFS restart after installing the branch.

### T1 Windmilling relight: engine 1 relights by the airflow alone, no APU

Setup: in flight at FL200, 300 kt, AP and A/THR ON, APU OFF, X BLEED selector AUTO (overhead AIR panel), ENG START selector NORM (pedestal, engine panel with the four ENG MASTER levers).

| # | Do this | Expected result |
|---|---|---|
| 1 | flyPad > Failures > ATA 72 > "Engine 1 flameout (crew relight possible)": activate it. | Amber ENG 1 FAIL; engine 1 FF 0, N3 and EGT fall. |
| 2 | ENG START selector: IGN START. ENG MASTER 1: OFF, wait 30 s. | ENG 1 SHUT DOWN replaces ENG 1 FAIL. |
| 3 | ENG MASTER 1: ON. | FF appears within a few seconds, engine 1 at idle within about 60 s, the alerts clear; SD ENGINE page: the engine 1 start valve stays closed. Lever in CL: engine 1 N3 holds at idle until the engine runs, then rises to the N3 of the other engines without passing them by more than 1-2 %. |
| 4 | ENG START selector NORM; flyPad: deactivate the failure. | - |

### T2 Quick relight: engine 4 relights at NORM

Setup: FL150, 240 kt, ENG START selector NORM.

| # | Do this | Expected result |
|---|---|---|
| 1 | Activate "Engine 4 flameout (crew relight possible)"; at once (within 5 s) ENG MASTER 4 OFF, then ON. | Engine 4 FF and EGT rise within a few seconds, back at idle, ENG 4 FAIL goes; the selector stays at NORM. No yaw kick when the relight ends (about 40 s later); about 10 s after that engine 4 joins the others with no overshoot. |
| 2 | flyPad: deactivate the failure. | - |

### T3 Starter assisted relight with crossbleed air: engine 3

Setup: FL150, 210 kt, APU OFF.

| # | Do this | Expected result |
|---|---|---|
| 1 | X BLEED selector OPEN. Activate "Engine 3 flameout (crew relight possible)". | ENG 3 FAIL. |
| 2 | ENG START selector IGN START; ENG MASTER 3 OFF, wait 30 s; ENG MASTER 3 ON. | SD ENGINE page: engine 3 start valve open, FF within a few seconds, engine 3 at idle within about 60 s, start valve closed again, ENG 3 FAIL goes. |
| 3 | ENG START NORM, X BLEED AUTO, deactivate the failure. | - |

### T4 On the ground: relight with APU air, and two engines out for taxi

Setup: on the ground, four engines at idle, parking brake ON, APU AVAIL, APU BLEED pb ON (overhead AIR panel), ENG START NORM.

| # | Do this | Expected result |
|---|---|---|

| | | |
|---|---|---|
| 1 | Activate "Engine 1 flameout (crew relight possible)"; when N1 is below 5 %: ENG MASTER 1 OFF, ENG START IGN START, wait 30 s, ENG MASTER 1 ON. | ENG 1 FAIL, then a normal start: start valve open, engine 1 at idle within about 60 s, ENG 1 FAIL goes. |
| 2 | ENG START NORM, deactivate the failure. ENG MASTER 1 OFF and ENG MASTER 4 OFF, wait 20 s. | No TWO ENGS OUT ON OPPOSITE SIDE caution, no ENG 1 / ENG 4 FAIL. |

### T5 Two engines out at 255 kt: engine 1 relights by windmilling

Setup: FL100, 255 kt (FCU speed), flaps 0, AP and A/THR ON, APU OFF, APU BLEED OFF, X BLEED AUTO. Keep at least 250 kt: descend if needed.

| # | Do this | Expected result |
|---|---|---|
| 1 | Activate "Engine 1 flameout (crew relight possible)", then "Engine 3 flameout (crew relight possible)". Wait 30 s. | ENG TWO ENGS OUT ON OPPOSITE SIDE with LAND ANSA; STATUS INOP SYS GEN 1+3. |
| 2 | ENG START selector IGN START. ENG MASTER 1 OFF, wait 30 s, ENG MASTER 1 ON. | Engine 1 lights up (fuel flow, EGT rise) and reaches idle; engine 3 stays out. SD ENGINE page: "A B" on engine 1 during the relight. |
| 3 | Deactivate both failures; relight engine 3 the same way; ENG START NORM. | - |

### T6 All engines out at FL200

Setup: FL200, 300 kt, APU OFF, ENG START selector NORM, four engines running.

| # | Do this | Expected result |
|---|---|---|
| 1 | Activate the four "Engine n flameout (crew relight possible)" failures one after the other. Wait 20 s. | Red ENG ALL ENG FLAME OUT; the RAT deploys; AC ESS powered by the emergency generator. With `core/a380x-elec` merged: no ELEC EMER CONFIG. |
| 2 | Press the RAT MAN ON pb (overhead EMER ELEC PWR panel, guarded). | The RAT MAN ON line turns green. |
| 3 | ENG START selector IGN START. ENG MASTER 4 OFF, wait 30 s, ENG MASTER 4 ON. | Engine 4 relights and ALL ENG FLAME OUT goes; the ENG FAIL alerts of the other engines follow. |
| 4 | Deactivate the failures and relight the other engines. | - |

> **NOTE**
> Recording used for the 2026-10-05 tests: per engine FUEL_CUT, RELIGHT_ATTEMPT, RELIGHT_IGNITION, ENGINE_STATE, ENGINE_N3, PNEU starter valve / pressurized, MSFS TURB ENG N2, GENERAL ENG COMBUSTION and fuel flow, with the airspeed, altitude, ENG START selector, X BLEED and APU BLEED.

## 8. Automatic relight with both igniters, and the engine feed tank fuel draw

> **Aircraft:** A380X | **Where:** Automatic (FADEC); pedestal ENG MASTER levers; E/WD | **Status:** Auto relight sim-tested OK on develop 2026-10-06 (relight batch T3, recorded: ignition 0.5 s after the combustion loss with the ENG START selector at NORM, relit when the fuel came back, ignition off 60 s later). Feed tank draw: passive check in the Flight A climb of 2026-10-06 (four engines kept burning), no dedicated reproduction. Native tests auto_relight_test (recorded flameout replay), feed_tank_draw_test. The core branch itself was not sim-tested.

### What it is for

In a flight of 2026-10-06 MSFS dropped the combustion of all four engines by itself at 7 900 ft. With the ENG START selector at NORM nothing relit them: the engines windmilled for 150 s until the crew selected IGN START. Two

changes:

- **AUTO RELIGHT** (A380 FCOM DSC-70-80-30-20 QUICK AND AUTO RELIGHT: "If the FADEC detects an engine flame out, on ground or in flight, continuous ignition with both igniters is automatically selected. The ignition is maintained for 60 s after engine relight."). When a running engine (ENG MASTER ON, FADEC supplied, fuel not cut) loses its combustion for 0.5 s, its FADEC turns on both igniters whatever the ENG START selector position, until 60 s of continuous combustion after the relight.
- **Feed tank draw.** Each engine is fed by MSFS through a 1 gallon buffer tank that the FADEC drains by the fuel burned. The FADEC took the whole burn of an update at once and compared kilograms with pounds, so one long sim frame could empty the four buffer tanks together and leave no fuel at any engine (the suspected cause of the four-engine flameout). Now, while the feed tank holds fuel, the FADEC leaves a small quantity in the buffer and carries the rest of the burn over to the next updates, so the fuel burned is still taken in full. A dry feed tank still empties the buffer and the engine flames out (fuel exhaustion).

## What you should see

| Situation | What you should see |
|---|---|
| **MSFS loses the combustion of a running engine, ENG START at NORM** | Within about 1 s the FADEC turns on the ignition; the engine burns again as soon as it has fuel, back at the thrust of the others within about 30 s. SD ENGINE page: "A B" on that engine. 60 s after the relight the ignition goes off. |
| **ENG MASTER OFF, ENG FIRE pb released, or a flyPad flameout / seizure failure** | No auto relight: a manual abort, an unpowered FADEC or a failure the crew must relight (section above). |
| **Long sim frame (stutter) in a climb** | The four engines keep burning; the fuel used rises steadily. |

## Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **There is no flyPad failure for an MSFS flameout.** | The flyPad flameout cuts the fuel on purpose and asks for a crew relight; the auto relight is for a flameout the FADEC did not cause. | The 2026-10-06 test closed the MSFS fuel valve of engine 4 for 5 s with a SimConnect tool. Since the HP fuel valve hold (below) the systems reopen an MSFS valve that disagrees with them, so that way of testing probably no longer cuts the fuel. |
| **A full MSFS restart is needed after an update.** | The FADEC WASM module changes. | Restart the sim after installing. |

## How to use

1. Nothing to set: keep the ENG MASTER levers ON and the ENG START selector at NORM in flight.
2. If an engine flames out by itself, watch it relight on the E/WD; follow ENG FAIL if it does not.

## How to test

The flameout needs MSFS to lose the combustion by itself, or a SimConnect tool that interrupts the engine fuel (see the limitation above).

| # | Do this | Expected result |
|---|---|---|
| 1 | A380X at FL150, 240 kt, flaps 0, four engines running, levers in CL, A/THR ON, ENG START selector NORM, ENG 4 ANTI ICE and WING ANTI ICE OFF (so the MSFS automatic ignition is off). | - |

| | | |
|---|---|---|
| 2 | Interrupt the fuel of engine 4 for 5 s (SimConnect tool). Touch nothing for 90 s. | Engine 4 N1 and N3 start to fall; within about 1 s of the fuel coming back it burns again with the selector still at NORM, and it is back at the thrust of the others within about 30 s. A brief ENG 4 FAIL is possible. |
| 3 | Climb from FL50 to FL100 at 250 kt, levers in CL, without touching the engine controls. | No ENG FAIL, no N1 / N3 drop; the FUEL USED rises steadily on the four engines. |

## 9. SD ENGINE page igniter indication

> **Aircraft:** A380X | **Where:** SD ENGINE page, start parameters of each engine | **Status:** Sim-tested OK on develop: 2026-10-06 (Flight A S2 T1 ground start with igniter A failed: "A", nothing during the crank, then "A B"; S3 T3 "A B" on all engines at IGN START in flight; S4 T2 "A B" during a relight) and 2026-10-07 (a second ground start of the same engine shows "B"). Native test igniter_selection_test, unit test StartParameters.spec.ts. The core branch itself was not sim-tested.

### What it is for

The SD ENGINE page drew an "A" from the N2 during a start, whatever the FADEC did. It now shows the igniters the FADEC really energizes (A380 FCOM DSC-70-90: "The igniter A(B) is energized." / "Both igniters A and B are energized."), in this priority order:

1. None with the ENG MASTER OFF or the FADEC unpowered (ENG FIRE pb released).
2. "A B" when the FADEC selects both igniters itself: auto relight, an in-flight relight that lights up.
3. "A B" for the quick relight (ENG MASTER OFF then ON within 30 s above 45 % N3), for 60 s.
4. Start: in flight "A B"; on the ground the igniters of the start sequence (next section): one igniter from 20 % to 58 % N3, A or B alternated at each start, "A B" for a new automatic attempt, none during the automatic dry crank.
5. Continuous ignition (ENG START at IGN START, engine running): "A B" at once in flight; on the ground only after the selector was set to NORM then back to IGN START after the start, and not at low power (N1 below 53 % for more than 30 s), as in FCOM DSC-70-80-20.

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **A failed igniter is still shown when it is energized ("A" during the first attempt with igniter A failed).** | The FCOM shows the igniter energized and has no line for a failed one; kept by decision (2026-10-08). | - |
| **The continuous ignition inhibition at high combustion chamber pressure is not modelled.** | MSFS has no combustion chamber pressure. | - |
| **A full MSFS restart is needed after an update.** | The FADEC WASM module changes. | Restart the sim after installing. |

### How to use

1. Open the SD ENGINE page (ECP ENG key); the start parameters show the igniter letters above the start valve of each engine during a start, a relight or the continuous ignition.

### How to test

| # | Do this | Expected result |
|---|---|---|

| | | |
|---|---|---|
| 1 | A380X on the ground, APU AVAIL, APU BLEED ON, ENG START selector IGN START, SD ENGINE page. ENG MASTER 2 ON. | A green "A" (or "B") above the engine 2 start valve from about 20 % N3 to about 58 % N3, then no letter; no letter below 20 % N3. |
| 2 | ENG MASTER 2 OFF; wait until N3 is below 10 %; ENG MASTER 2 ON. | The other letter, in the same way (A, then B). |
| 3 | In flight at FL100, all engines running, ENG START selector NORM. Set IGN START, wait 5 s. | On all four engines the start parameters replace the nacelle temperatures and show "A B". |
| 4 | Set NORM, wait 5 s. | The letters go out and the nacelle temperatures come back. |

## 10. Engine ignition and start failures, FCOM ground start

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 74 Ignition and ATA 80 Starting; ENG START selector, ENG MASTER levers, ENG MAN START pbs; E/WD, STATUS, SD ENGINE page | **Status:** Sim-tested OK on develop 2026-10-06 (Flight A S2 T1, recorded): dry crank with the ENG MAN START pb, then "Engine 1 igniter A": no light up, START FAULT NO LIGHT UP, automatic crank, light up at the second attempt. The other start failures (start valve, hot, hung, stall, starter) are not sim-tested. Rust tests (engine_start.rs, a380_systems engine_failure.rs), native test start_sequence_test, unit tests EngineStartAlerts.spec.ts, a380.spec.ts. The core branch itself was not sim-tested.

### What it is for

The FBW FADEC ran the engine model but not what a real FADEC does during a start. The ground start now follows the A380 FCOM (DSC-70-80-20 IGNITION, DSC-70-80-30 STARTING, DSC-70-80-40 CRANKING): ENG START selector to IGN START, then ENG MASTER ON; one igniter (alternated between starts) and fuel at 20 % N3; start valve closed and ignition off at about 58 % N3. A start that does not light up within 20 s, stalls, hangs or goes above 745 °C EGT is aborted, dry cranked for 30 s and tried again with both igniters, up to three attempts. A start valve stuck closed, no starter air or a failed starter leaves the core at rest. The ENG MAN START pbs work: dry crank (ENG START at CRANK, MAN START ON, ENG MASTER OFF), wet crank and manual start.

| flyPad failure (ATA) | What it does |
|---|---|
| Engine 1 (2)(3)(4) igniter A / igniter B (74) | That igniter does not light the engine. The first automatic attempt with only that igniter does not light up: START FAULT NO LIGHT UP, then a new attempt with both. ENG n IGN A (B) FAULT; both failed: ENG n IGN A+B FAULT. |
| Engine 1 (2)(3)(4) start valve stuck closed (80) | No starter air: the core stays at rest. ENG n START VLV FAULT (NOT OPEN) with VLV STUCK CLOSED. |
| Engine 1 (2)(3)(4) start valve stuck open (80) | The valve does not close after the start: ENG n START VLV FAULT (NOT CLOSED). |
| Engine 1 (2)(3)(4) hot start (start EGT over limit) (80) | The start EGT overshoots: START FAULT EGT OVERLIMIT, the start is aborted. |
| Engine 1 (2)(3)(4) hung start (N3 stops below idle) (80) | The core stops accelerating below idle: START FAULT HUNG START. |
| Engine 1 (2)(3)(4) stall during the start (80) | START FAULT ENG STALL. |
| Engine 1 (2)(3)(4) starter shaft shear (80) | The starter does not turn the core although it gets air: START FAULT LOW N2, no further attempt. |

## ECAM alerts

| Alert | Lines (A380 FCOM PRO-ABN-ECAM-10-70) |
|---|---|
| ENG n START FAULT | The sub-case (NO STARTER AIR PRESSURE, LOW N2, HUNG START, EGT OVERLIMIT, NO LIGHT UP, ENG STALL, STARTER TIME EXCEEDED, THR LEVERS NOT AT IDLE), then: no starter air: XBLEED OPEN, APU BLEED ON (in flight MIN SPEED FOR WINDML RELIGHT : 260 KT); automatic start on the ground: NEW START IN PROGRESS, then ENG n MASTER OFF once aborted; manual start: ENG n MASTER OFF, ENG START SEL CRANK, AFTER 30S: ENG START SEL NORM, ENG n MAN START OFF; levers not at idle: ALL THR LEVERS IDLE. |
| ENG n START VLV FAULT (NOT CLOSED) | ENG n BLEED OFF (in flight), APU BLEED OFF (engine 1), X BLEED CLOSE, ENG n MAN START OFF (manual start), ENG n MASTER OFF (on the ground), WING A-ICE OFF and AVOID ICING CONDs (engines 1 and 4). |
| ENG n START VLV FAULT (NOT OPEN) | VLV STUCK CLOSED; on the ground ENG n MAN START OFF, ENG n MASTER OFF; in flight MIN SPEED FOR WINDML RELIGHT : 260 KT, WINDMILL START INITIATED, ENG PARAMETERS MONITOR WITH CARE. |
| ENG n IGN A (B) FAULT | STATUS MORE: REDUND LOSS ENG n IGN A (B). |
| ENG n IGN A+B FAULT | STATUS INOP SYS ENG n IGNITION. |

> **NOTE**
> The start alerts are inhibited in the flight phases 3 to 7, 9 and 10 (IGN A (B) FAULT also in phase 8), per the FCOM figures. START FAULT and IGN A (B) FAULT come without chime and MASTER CAUT; IGN A+B FAULT keeps them (see "Engine alerts as in the FCOM" below for the in-flight START FAULT cases).

## Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **Hung start, stall and starter detection times are design values.** | The FCOM gives the symptoms, not the times; the FBW engine model has no compressor pressure. | - |
| **START FAULT LOW N1 is not modelled.** | No N1 rotor failure. | - |
| **A full MSFS restart is needed after an update.** | The systems and FADEC WASM modules and the pedestal behaviour XML change. | Restart the sim after installing. |

## How to use

1. Normal start: APU BLEED ON, ENG START selector IGN START, ENG MASTER ON; the FADEC does the rest (ENG START NORM when the engines are running).
2. Failures: flyPad > Failures > ATA 74 Ignition or ATA 80 Starting; activate one, start the engine and follow the ECAM.
3. Dry crank: ENG START selector CRANK, ENG MAN START n pb ON (ENG MASTER OFF), then OFF.

## How to test

From S2 T1 of `test-plan-integration-2026-10-06.md`. Setup: A380X cold and dark, batteries ON, APU AVAIL, APU BLEED ON, all ENG MASTER levers OFF, thrust levers at IDLE, SD ENGINE page.

| # | Do this | Expected result |
|---|---|---|
| 1 | ENG START selector CRANK. ENG MAN START 2 pb (overhead ENG MAN START panel) ON, wait 20 s, then OFF. | Engine 2 N3 rises to about 20-25 % and stays there; no fuel flow, no EGT rise; it does not start. After MAN START OFF its N3 runs down. No ENG 2 START FAULT. |

| | | |
|---|---|---|
| 2 | flyPad > Failures > ATA 74 Ignition > "Engine 1 igniter A": activate. ENG START selector IGN START. ENG MASTER 1 ON, wait 80 s. | About 22 s after MASTER ON: amber ENG 1 START FAULT with NO LIGHT UP and NEW START IN PROGRESS, and amber ENG 1 IGN A FAULT; N3 at 20-30 %, no EGT, no fuel flow. SD: "A" during the first attempt, nothing during the crank, "A B" for the second attempt. About 55 s after MASTER ON: EGT rises, engine 1 starts, START FAULT goes. |
| 3 | Deactivate the failure. Activate "Engine 3 start valve stuck closed"; ENG MASTER 3 ON. | Engine 3 stays at rest; ENG 3 START VLV FAULT (NOT OPEN) with VLV STUCK CLOSED. ENG MASTER 3 OFF, deactivate. |
| 4 | Activate "Engine 3 hot start (start EGT over limit)"; ENG MASTER 3 ON. | The EGT overshoots: ENG 3 START FAULT EGT OVERLIMIT, the start is aborted and cranked, then a new attempt. ENG MASTER 3 OFF, deactivate. |

## 11. HP fuel valve: no engine burning without a FADEC start

> **Aircraft:** A380X | **Where:** Pedestal ENG MASTER levers and ENG START selector | **Status:** Sim-tested on develop: the fault found 2026-10-06 (Flight A S2 T3, a restart 34 s after a shutdown skipped the start sequence); 2026-10-07 restart after shutdown OK (full start sequence, igniter B), ENG MASTER ON at NORM stays dark, a start at IGN START opens the HP valve at the fuel-on point; 2026-10-07 evening the HP valves stayed closed from load through power-up (more than 200 s). Rust tests (engine_failure.rs HpFuelValve, sim_fuel_valve.rs), native start_sequence tests. The core branch itself was not sim-tested.

### What it is for

A380 FCOM DSC-70-30 ENGINE SHUTDOWN: with the ENG MASTER OFF "The FADEC closes the LP and HP fuel valves. The engine decelerates and stops." The ENG MASTER only closed an MSFS valve upstream of the engine buffer tank, so MSFS kept burning a shut down engine for 20 to 60 s, and a restart in that time skipped the start sequence. And with the ENG START selector at NORM, ENG MASTER ON on the ground let the MSFS starter turn the core and MSFS lit it at 22 % N2 without any FADEC start (the FCOM: IGN START first, fuel at 20 %; at NORM the MASTER ON starts nothing).

- The HP fuel valve of each engine (the MSFS fuel valves 60-63 after the buffer tanks) is now closed with the ENG MASTER OFF or the fuel cut, and on the ground until the FADEC start sequence reaches its fuel-on point (IGN START then MASTER ON) or the quick relight opens it (MASTER OFF then ON within 30 s above 45 % N3). It is open for the aircraft presets and a flight loaded with the master ON. In flight the relight logic keeps the fuel cut until the engine may light up. Once open it stays open until the MASTER OFF or a fuel cut, so a running engine keeps its fuel at NORM and its auto relight has fuel.
- MSFS could reopen a closed valve by itself a few seconds later (flight load, power-up of the valve circuit). The systems now watch the MSFS valve switch and command it again whenever it disagrees, and once on the first update of a flight.
- An engine is shown running (ON) only when MSFS really burns in it: a core turned past 20 % by the MSFS starter at NORM no longer shows ON unlit (and no longer triggers the auto relight).

### What you should see

| Situation | What you should see |
|---|---|
| **ENG MASTER OFF** | The engine stops burning at once (FF 0, EGT falls) and runs down. |
| **ENG MASTER ON again soon after a shutdown, ENG START at IGN START** | A full start sequence: start valve, one igniter, fuel at 20 % N3. |
| **ENG MASTER ON on the ground with the ENG START selector at NORM** | Nothing lights up: no fuel flow, no EGT. |

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **A full MSFS restart is needed after an update.** | The systems WASM module changes. | Restart the sim after installing. |

### How to use

1. Start an engine the FCOM way: ENG START selector IGN START first, then ENG MASTER ON.
2. Shut it down with ENG MASTER OFF.

### How to test

| # | Do this | Expected result |
|---|---|---|
| 1 | A380X cold and dark on the ground, APU AVAIL, APU BLEED ON, ENG START selector NORM. ENG MASTER 1 ON, wait 60 s. | Engine 1 stays dark (no fuel flow, no EGT), also after the aircraft power-up. |
| 2 | ENG MASTER 1 OFF. ENG START selector IGN START, ENG MASTER 1 ON. | A normal start: fuel at about 20 % N3, engine 1 at idle. |
| 3 | ENG MASTER 1 OFF; after about 30 s (engine still turning) ENG MASTER 1 ON. | Fuel flow goes to 0 at once at the MASTER OFF; the restart goes through the start sequence again (start valve opens, one igniter on the SD). |

## 12. Engine oil failures (leak, filter clog, overheat)

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 79 Oil; E/WD; SD ENGINE page | **Status:** Sim-tested on develop 2026-10-06 (Flight A S5 T2): filter clog and leak to OIL PRESS LO and the shutdown OK; the overheat stopped at 128 °C (fixed: the FADEC keeps its own temperature). 2026-10-07: the overheat reached the 196 °C limit only after 4 min (then retuned), the cool-down after clearing the failure OK. The retuned overheat curve (OIL TEMP HI about 2 min at cruise) and the oil pressure from the whole oil system are not sim-tested on the A380X (the latter checked on the A32NX 2026-10-07). Native tests engine_oil_failures_test, oil_system_test, unit tests OilIndications.spec.ts, EngineOilAlerts.spec.ts. The core branch itself was not sim-tested.

### What it is for

Twelve new failures, three per engine, computed by the FADEC on the oil quantity, pressure and temperature, with the SD ENGINE page indications and the ECAM alerts of the A380 FCOM (DSC-70-90, PRO-ABN-ECAM-10-70).

| flyPad failure (ATA 79 Oil) | What it does |
|---|---|
| Engine 1 (2)(3)(4) oil leak | The oil system empties at 2 qt/min (the tank reading falls about 1.6 qt/min); the pump pressure stays full down to 1.3 qt in the system and is gone at 0.3 qt, so OIL PRESS LO follows the low oil quantity indication. |
| Engine 1 (2)(3)(4) oil filter clog | CLOGGED on the SD ENGINE page and ENG n OIL FILTER CLOGGED; no other effect. |
| Engine 1 (2)(3)(4) oil overheat | The oil temperature rises towards a target that grows with the core speed (170 °C at idle, 230 °C at 100 % N3, 60 s time constant): OIL TEMP HI about 2 min after the failure at cruise; reducing thrust brings it below 196 °C, at IDLE within about 1 min. Once cleared, the oil cools down gradually. |

### What you should see

| Situation | What you should see |
|---|---|
| **SD ENGINE page, oil quantity** | Pulses below the 1.2 qt advisory (first white dash). |
| **SD ENGINE page, oil temperature** | Pulses above 163 °C. |

| | |
|---|---|
| **SD ENGINE page, oil pressure** | FCOM scale 0-100 / 100-440 PSI, red at or below 25 PSI; CLOGGED below it for a clogged filter. |
| **Oil pressure below 25 PSI** | Red **ENG n OIL PRESS LO** (MASTER WARN, continuous chime): THR LEVER n IDLE, ENG n MASTER OFF; then ENG n SHUT DOWN. |
| **Oil temperature above 196 °C** | Amber **ENG n OIL TEMP HI**: THR LEVER n REDUCE BELOW OIL TEMP LIM, IF HI TEMP PERSISTS: ENG n MASTER OFF. |
| **Clogged filter** | Amber **ENG n OIL FILTER CLOGGED** on the ground (phases 1, 2, 11 and 12), without chime or MASTER CAUT; in flight only the SD CLOGGED. |

## Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **The leak and overheat rates are design values.** | The FCOM gives no rates. | - |
| **The alerts are monitored only with the engine running and its ENG MASTER ON.** | Design choice: a stopped engine has no oil pressure, and the procedures end with the MASTER OFF. | - |
| **Flight phase inhibitions.** | OIL PRESS LO: phases 1, 5, 6, 12; OIL TEMP HI: 1, 4-7, 9, 10, 12; OIL FILTER CLOGGED: 3 to 10 (FCOM figures). | Test in cruise. |
| **A full MSFS restart is needed after an update.** | The systems and FADEC WASM modules change. | Restart the sim after installing. |

## How to use

1. Open flyPad > Failures > ATA 79 Oil and activate a leak, a filter clog or an overheat.
2. Watch the SD ENGINE page; follow the ECAM (reduce thrust, or IDLE and ENG MASTER OFF).
3. Deactivate the failure.

## How to test

From S5 T2 of `test-plan-integration-2026-10-06.md`. Setup: level at FL200, 300 kt, A/THR ON, four engines running, SD ENGINE page.

| # | Do this | Expected result |
|---|---|---|
| 1 | Activate "Engine 2 oil filter clog", "Engine 4 oil overheat" and "Engine 1 oil leak". | SD: amber CLOGGED under the engine 2 oil pressure (no E/WD alert in flight). Engine 4 oil temperature rises and pulses above 163; then amber ENG 4 OIL TEMP HI with THR LEVER 4 ... REDUCE BELOW OIL TEMP LIM. |
| 2 | When ENG 4 OIL TEMP HI shows: A/THR off, THR LEVER 4 IDLE; fly the speed with the other levers. | Engine 4 oil temperature falls below 196 and the alert goes within about 1-2 minutes. |
| 3 | Wait for engine 1 (several minutes). | Engine 1 oil quantity falls and pulses below 1.2 qt; then its oil pressure falls and turns red at 25 PSI: red ENG 1 OIL PRESS LO with MASTER WARN, THR LEVER 1 ... IDLE, ENG 1 MASTER ... OFF. |
| 4 | THR LEVER 1 IDLE, ENG MASTER 1 OFF. Deactivate the three failures. | OIL PRESS LO goes, ENG 1 SHUT DOWN comes. Engine 4 oil cools down gradually. |

## 13. Engine fuel filter clog, ENG MASTER FAULT light, SD start valve and N3 red cross

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 73 and ATA 80; pedestal ENG MASTER levers; SD ENGINE page; E/WD | **Status:** Sim-tested by the user on develop, data-checked OK (2026-10-08): fuel filter clog on engine 3 at idle; start valve stuck closed (START VALVE FAULT 5 s after ENG MASTER ON); starter shaft shear (start aborted and ENG 1 FAULT light 9.8 s after ENG MASTER ON, N3 at most 0.2 %). The N3 red cross is unit-tested only (not reachable on the ground).

### What it is for

- **Fuel filter clog:** new failures Engine 1-4 fuel filter clog (ATA 73). Amber **CLOGGED** below the fuel flow on the SD ENGINE page while the engine runs, and the **ENG n FUEL FILTER CLOGGED** caution (FCOM: pressure drop above 35 PSI, crew awareness, inhibited in flight phases 3 to 10). No effect on the engine: the filter is bypassed.
- **ENG MASTER FAULT light:** it comes on when the FADEC aborts an automatic start (FCOM DSC-70-90), and goes off with the ENG MASTER OFF. It followed the MSFS engine failure before, which a start abort does not set.
- **SD start valve:** amber when the start valve is abnormally open or closed (START VLV FAULT); with a valve fault the start parameters show even at NORM, so a valve stuck open after the start is seen.
- **N3 red cross:** above the 118.7 % red limit a red "+" shows next to N3 until the next start of that engine on the ground (FCOM: N2 of the GP7200 core, read on the Trent N3).
- **Starter failure fix:** with a sheared starter the MSFS starter could still turn the core to about 6 %, past the 5 % starter fault detection, so the start never aborted. The FADEC now keeps the core at rest and aborts the start without a new attempt (FCOM DSC-70-80-30-20).

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **The FAULT light has no HP fuel valve and no overthrust cause.** | FBW has no HP fuel valve failure (the valve always follows its command) and no overthrust protection. | - |
| **A full MSFS restart is needed after an update.** | The systems and FADEC WASM modules and the cockpit behaviour change. | Restart the sim after installing. |

### How to use

1. Call the SD ENGINE page (ECP ENG key).
2. Open flyPad > Failures > ATA 73 (fuel filter clog) or ATA 80 (starter shaft shear, start valve stuck) and activate the failure, then start or run the engine as below.
3. Deactivate the failures after the test.

### How to test

From `reports/test-card-engine-leftovers-2026-10-08.md` (recorded OK on develop 2026-10-08). Setup: at the gate, APU running, APU BLEED ON, all ENG MASTER levers OFF, ENG START selector IGN START.

| # | Do this | Expected result |
|---|---|---|
| 1 | flyPad ATA 80 **Engine 1 starter shaft shear** ON; ENG MASTER 1 ON, wait 30 s; then OFF. | N3 stays near 0; about 10 s after the MASTER ON the start aborts: E/WD ENG 1 START FAULT and the amber FAULT light in the ENG 1 MASTER lever (the other three stay off); the light goes off with the MASTER OFF. |
| 2 | flyPad ATA 80 **Engine 2 start valve stuck closed** ON; ENG MASTER 2 ON, wait 15 s; then OFF. | The engine 2 start valve circle and bar are amber on the SD; E/WD ENG 2 START VLV FAULT. |

| | | |
|---|---|---|
| 3 | ENG MASTER 3 ON, wait until idle; flyPad ATA 73 **Engine 3 fuel filter clog** ON. | Amber CLOGGED below the engine 3 fuel flow; E/WD amber ENG 3 FUEL FILTER CLOGGED, without chime or MASTER CAUT (FCOM level, since 2026-10-12); engine 3 keeps running. |

## 14. Engine stall, EGT overtemperature and vibration failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 72 Engine and ATA 77 Engine Indicating; E/WD; SD ENGINE page | **Status:** Sim-tested OK on develop 2026-10-06 (Flight A S5 T1: engine 3 stall, engine 2 EGT overtemperature and high vibration). The retuned vibration (square of the rotor speed, 2026-10-07) is not sim-tested on the A380X. EWD EGT amber limit from the thrust limit: tested on develop 2026-10-04. Rust tests (engine_malfunction.rs), native test egt_failure_offset_test, unit tests EngineParameterAlerts, EgtLimits, EngineVibration specs. The core branch itself was not sim-tested.

### What it is for

Twelve new failures where the engine keeps running. What each one does is a design choice: the FCOM gives the symptoms, not values.

| flyPad failure (ATA) | What it does |
|---|---|
| Engine 1 (2)(3)(4) compressor stall (above 60 % N1) (72) | Above 60 % N1 command the engine stalls: it loses thrust (the thrust loop aims lower), the EGT rises quickly, N1 / N3 fluctuate. It recovers with the lever at idle and stalls again above the threshold. |
| Engine 1 (2)(3)(4) EGT overtemperature (72) | The EGT is higher, more so at high thrust: ENG EGT OVER LIMIT at climb thrust. |
| Engine 1 (2)(3)(4) high vibration (77) | N1, N2 and N3 vibrations growing with the square of the rotor speed: above the SD advisory (5 units) at climb and cruise thrust, below it at reduced thrust and at idle. At a low cruise power the advisory can show on N3 only (fan 60 %: N1 4.7; core 81 %: N3 5.3), as the FCOM advisory is per rotor; reducing thrust clears it. No ECAM alert (ABN PROC). |

### What you should see

| Situation | What you should see |
|---|---|
| Stall | Amber **ENG n STALL** (SD ENGINE page): THR LEVER n IDLE, ENG n PARAMETERS CHECK, IF ABNORMAL: ENG n MASTER OFF, ENG n RELIGHT PROC CONSIDER. Inhibited from V1 to 400 ft unless two or more engines stall. |
| EGT above its limit | Amber **ENG n EGT OVER LIMIT**: THR LEVER n REDUCE BELOW EGT LIMIT, IF OVER LIMIT PERSISTS: ENG n MASTER OFF; the amber attention box on the engine. |
| N1 or N2 above the red limit | Red **ENG n N1/N2 OVER LIMIT**: THR LEVER n REDUCE BELOW N1 LIMIT, IF OVER LIMIT PERSISTS: ENG n MASTER OFF. |
| E/WD EGT gauge | A red mark at the highest EGT reached above the red line. The amber limit comes from the active thrust limit (none at FLEX / TOGA or in reverse, the CLB / MCT limit otherwise), not from the lever angle. |
| SD ENGINE page | Separate N1, N2 and N3 vibrations that pulse above 5 units; N3 red above its limit. |

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|

| | | |
|---|---|---|
| **The A380X has no overspeed failure (the A32NX has one).** | Not part of this stage. The shared overspeed ceiling fields added for the A32NX LEAP-1A red lines (2026-10-12) are set to none on the A380X: no change here. | - |
| **A full MSFS restart is needed after an update.** | The systems, FADEC and fbw WASM modules change. | Restart the sim after installing. |

### How to use

1. Open flyPad > Failures > ATA 72 Engine (stall, EGT) or ATA 77 Engine Indicating (vibration) and activate the failure at climb or cruise thrust.
2. Follow the ECAM: reduce the thrust of that engine; deactivate the failure.

### How to test

From S5 T1 of `test-plan-integration-2026-10-06.md`. Setup: four engines running, A/THR ON, CLB, climbing at 280 kt.

| # | Do this | Expected result |
|---|---|---|
| 1 | Climbing through FL150: activate "Engine 3 compressor stall (above 60 % N1)", wait 20 s. | Amber ENG 3 STALL with its lines; SD ENGINE page shown; engine 3 THR / N1 drops and fluctuates, EGT 3 rises. |
| 2 | Thrust lever 3 IDLE, wait 20 s; deactivate the stall; lever 3 back to CL. | The stall alert goes. |
| 3 | Activate "Engine 2 EGT overtemperature" and "Engine 2 high vibration", wait 30 s. | EGT 2 above 850 °C, amber, with amber ENG 2 EGT OVER LIMIT and the attention box on engine 2; SD VIB N1, N2 and N3 of engine 2 above 5, pulsing. |
| 4 | Deactivate both. | The EGT and vibrations come back to normal; the alert goes. |

## 15. FADEC, thrust lever and reverser failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 73 Engine - Fuel And Control and ATA 78 Exhaust; thrust levers; E/WD, STATUS, SD ENGINE page | **Status:** Sim-tested on develop 2026-10-06 (Flight A S4 T1): "Engine 1 FADEC network link" alerts and displays OK but the engine stayed at its thrust (fixed: the engine now follows its lever, not re-tested in the sim); "Reverser 2 unlocked (engine at idle)" OK. Reverse thrust OK 2026-10-07 (engines 2 and 3). The other FADEC, thrust lever and reverser failures and ENG REVERSER SELECTED are not sim-tested. Rust tests (engine_control_failure.rs, reverser), native tests (fadec_failure_inputs_test, fadec_fault_manual_thrust_test), unit tests FadecReverserAlerts.spec.ts, EngineParametersAvailability.spec.ts. The core branch itself was not sim-tested.

### What it is for

Twenty-six new failures with the alerts, procedures and STATUS of the A380 FCOM (PRO-ABN-ECAM-10-70). Reversers exist on engines 2 and 3 only.

| flyPad failure (ATA) | What it does | ECAM |
|---|---|---|
| Engine 1 (2)(3)(4) FADEC network link (73) | The FADEC still controls the engine, but the A/THR and the SD ENGINE page lose it (XX on the SD; the E/WD keeps N1, THR, EGT). The engine follows its thrust lever. | ENG n FADEC FAULT: THR LEVER n MAN ADJUST, ENG n PARAMETERS MONITOR, IF ABNORMAL: THR LEVER n IDLE, ENG n MASTER OFF; STATUS INOP SYS ENG n A/THR (A/THR engaged). |

| | | |
|---|---|---|
| Engine 1 (2)(3)(4) FADEC system fault (73) | Alert only (crew awareness). | ENG n FADEC SYS FAULT |
| Engine 1 (2)(3)(4) FADEC overheat (73) | Alert only (crew awareness). | ENG n FADEC TEMP HI |
| Thrust lever 1 (2)(3)(4) resolvers (both) (73) | The FADEC has lost the lever: the engine is at idle on the ground and limited to CLB in flight (the A/THR still manages it between idle and CLB). | ENG n THR LEVER FAULT: on the ground ENG n IDLE ONLY, THR LEVER n IDLE; in flight ENG n CLB ONLY, THR LEVER n CLB; STATUS INOP SYS ENG n THR LEVER, INFO ENG n IDLE (CLB) ONLY. |
| Reverser 2 (3) fault (does not deploy) (78) | The reverser does not deploy; the engine stays at reverse idle. | ENG 2 (3) REVERSER FAULT: T.O PERF CHECK (on the ground), LDG DIST IMPACT ON WET/CONTAM RWY ONLY. |
| Reverser 2 (3) unlocked (engine at idle) (78) | The cowls stay stowed (no reverse thrust, no buffet); the FADEC sets the engine at idle. | ENG 2 (3) REVERSER UNLOCKED: ENG n IDLE ONLY, THR LEVER n IDLE, ENG n MASTER OFF on the ground; in flight IF BUFFET: ENG n MASTER OFF; STATUS INFO ENG n IDLE ONLY. |
| Reverser 2 (3) energized (tertiary lock unlocked) (78) | The tertiary lock stays released; the primary locks hold the cowls stowed. | ENG 2 (3) REVERSER ENERGIZED: THR LEVER n IDLE. |
| Reverser 2 (3) failed locked (does not deploy) (78) | The locks are not released: no deployment. | ENG 2 (3) REV LOCKED: T.O PERF CHECK, LDG DIST IMPACT ON WET/CONTAM RWY ONLY. |
| Reverser 2 (3) control fault (ETRAC) (78) | The reverser is inoperative. | ENG 2 (3) REVERSER CTL FAULT: T.O PERF CHECK, LDG DIST IMPACT ON WET/CONTAM RWY ONLY. |

## Also in this feature

- **Reverse thrust stays at reverse idle until the reverser is fully deployed** (FCOM DSC-70-70): a reverser that does not deploy no longer gives forward thrust with the lever in reverse.
- **Reverse thrust reaches MSFS.** The in-flight start throttle hold (engine failures section) clamped the MSFS throttle at 0 and so blocked all reverse thrust; the reverse throttle now goes through.
- **An engine with a FADEC FAULT follows its thrust lever.** In the 2026-10-06 test the engine stayed at its N1 with the levers in CL: the FADEC THRUST LOCK latched when the lost link took the A/THR away. The A380 ENG FADEC FAULT does not disconnect the A/THR (only that engine is lost, INOP SYS ENG n A/THR), and the procedure says THR LEVER ... MAN ADJUST, so the lock no longer latches for it.
- **ENG REVERSER SELECTED** comes up when the lever of engine 2 or 3 is in reverse in flight: THR LEVER 2 (3)(2+3) FWD THR.

## Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **ENG 2 (3) REVERSER INHIBITED and REVERSER MINOR FAULT are not wired.** | No flyPad failure for them (maintenance action / no effect). | - |
| **"Before takeoff" lines (T.O PERF CHECK) show on the ground.** | Design choice: before takeoff = on the ground. | - |

| | | |
|---|---|---|
| **A full MSFS restart is needed after an update.** | The systems and fbw WASM modules change. | Restart the sim after installing. |

### How to use

1. Open flyPad > Failures > ATA 73 Engine - Fuel And Control (FADEC, thrust lever) or ATA 78 Exhaust (reversers 2 and 3) and activate a failure.
2. Follow the ECAM; for a FADEC FAULT adjust that thrust lever by hand.
3. Deactivate the failure.

### How to test

From S4 T1 of `test-plan-integration-2026-10-06.md` and the 2026-10-07 retest. Setup: FL100, 250 kt, flaps 0, AP and A/THR ON, four levers in CL.

| # | Do this | Expected result |
|---|---|---|
| 1 | Activate "Engine 1 FADEC network link". | Amber ENG 1 FADEC FAULT with THR LEVER 1 .... MAN ADJUST; SD ENGINE page XX for engine 1 while the E/WD keeps its N1 / THR / EGT; STATUS INOP SYS ENG 1 A/THR. Engine 1 goes to the CLB thrust of its lever; the A/THR still manages engines 2 to 4. |
| 2 | Move thrust lever 1 out of CL and back. | Engine 1 follows its lever (it does not stay frozen). |
| 3 | Activate "Reverser 2 unlocked (engine at idle)". | Amber ENG 2 REVERSER UNLOCKED with ENG 2 IDLE ONLY, THR LEVER 2 .... IDLE, .IF BUFFET, ENG 2 MASTER .... OFF; engine 2 N1 goes to idle; STATUS INFO ENG 2 IDLE ONLY. |
| 4 | Deactivate both. Activate "Thrust lever 4 resolvers (both)". | ENG 4 THR LEVER FAULT with ENG 4 CLB ONLY, THR LEVER 4 CLB; engine 4 limited to CLB. Deactivate it. |
| 5 | After landing: all thrust levers to full reverse until 70 kt, then IDLE. | REV on engines 2 and 3 only; their N1 rises above idle once the reversers are deployed. |

## 16. ENG THR LEVERS NOT SET and ENG OIL TEMP LO

> **Aircraft:** A380X | **Where:** E/WD; thrust levers; T.O CONFIG pb; MFD FMS PERF T.O (FLEX) | **Status:** THR LEVERS NOT SET: tested in the sim by the user: OK, data-confirmed (2026-10-08, on the feature branch). OIL TEMP LO: not sim-tested. Unit tests EngineTakeoffAlerts.spec.ts.

### What it is for

Two take-off alerts of the A380 FCOM (PRO-ABN-ECAM-10-70) that the A380X did not trigger. Both are amber cautions (MASTER CAUT, single chime), shown in flight phases 2 and 3 only.

- **ENG THR LEVERS NOT SET** (l.175146-175184): a thrust lever of a running engine is at or between CL and FLX/MCT, or at FLX/MCT without a FLEX TEMP, for 3 s. Line: ALL THR LEVERS ... TOGA (no FLEX TEMP: TOGA take-off) or THR LEVERS ... MCT/FLEX (FLEX TEMP entered, levers below FLX/MCT).
- **ENG 1(2)(3)(4) OIL TEMP LO** (l.172605-172628): on the ground the engine oil is below 50 °C 30 s after the engine is running, or after the T.O CONFIG pb, or at take-off power. Lines: THR LEVER n ... IDLE, DELAY T.O FOR WARM UP.

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|

| | | |
|---|---|---|
| **No derated take-off case.** | FBW has no derated take-off. | - |
| **A T.O CONFIG press keeps the oil monitor armed until the engine stops.** | Design choice: the FCOM says "after the flight crew presses". | - |

### How to use

1. Nothing to set. Set all thrust levers to FLX/MCT (FLEX TEMP entered) or TOGA for take-off.
2. After an engine start in the cold, wait until the oil is warm before take-off.

### How to test

| # | Do this | Expected result |
|---|---|---|
| 1 | A380X on the ground, PARKING BRK ON, engine 4 running at idle, all levers at IDLE, no FLEX TEMP. Thrust lever 4 only to the CL detent. | After 3 s: amber ENG THR LEVERS NOT SET with ALL THR LEVERS ... TOGA, MASTER CAUT. |
| 2 | Lever 4 back to IDLE. | The caution goes. |
| 3 | Cold and dark at a cold airport (oil at the outside temperature, below 50 °C): start engine 1 and wait 30 s. | Amber ENG 1 OIL TEMP LO with THR LEVER 1 ... IDLE and DELAY T.O FOR WARM UP while the oil is below 50 °C (SD ENGINE page). |

## 17. FADEC thrust failures: ENG THRUST LOSS and ENG T.O THRUST DISAGREE

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 73; E/WD; thrust levers | **Status:** Tested in the sim by the user on the feature branch: T.O THRUST DISAGREE OK (2026-10-08, FADEC 4 take-off mode recorded); THRUST LOSS OK with levers 3 and 4 at FLX/MCT (2026-10-08, max THR 85 recorded, the caution seen). Unit tests EngineThrustAlerts.spec.ts, a380.spec.ts; native test fadec_thrust_failures_test.cpp.

### What it is for

Two FCOM cautions that need a FADEC fault the sim cannot produce, with the failures that make them possible (they act on the FADEC models of the fbw WASM):

- New flyPad failures, ATA 73: **Engine 1 (2, 3, 4) max thrust miscalculated (thrust loss)** (73050-73053): that FADEC computes a maximum THR 15 % lower, so the engine gives less thrust above idle (N1 command = IDLE + 0.85 x (limit - IDLE)).
- **ENG 1(2)(3)(4) THRUST LOSS** (amber, l.173378-173408, phase 3 only): the maximum THR of an engine is 10 % lower than that of at least two other engines. Lines: THR LEVER n ... IDLE, ENG n MASTER ... OFF.
- New flyPad failures, ATA 73: **FADEC 1 (2, 3, 4) FLEX TEMP not received (T.O mode TOGA)** (73060-73063): that FADEC takes off in TOGA mode (MCT thrust in the FLX/MCT detent).
- **ENG T.O THRUST DISAGREE** (amber, l.175054-175070, phases 2 and 11): the FADECs do not all have the same take-off mode. No line (crew awareness).

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **THRUST LOSS shows in phase 3 only.** | FCOM inhibition; the A380X FWS needs two levers at take-off power for phase 3. | Set two levers (for example 3 and 4) to FLX/MCT. |
| **Use FADEC 2, 3 or 4 for the FLEX TEMP failure.** | The E/WD thrust rating is the one of FADEC 1. | - |

| | | |
|---|---|---|
| **A full MSFS restart is needed after an update.** | The fbw WASM module changes. | Restart the sim after installing. |

### How to use

1. flyPad > Failures > ATA 73 Engine - Fuel And Control: activate one of the failures before take-off.
2. Follow the caution; clear the failure to end it.

### How to test

| # | Do this | Expected result |
|---|---|---|
| 1 | A380X on the ground, PARKING BRK ON, engines running at idle, all levers at IDLE, FLEX TEMP 60 on the MFD FMS ACTIVE/PERF T.O page. flyPad > Failures > ATA 73: activate **FADEC 4 FLEX TEMP not received (T.O mode TOGA)**. | Amber ENG T.O THRUST DISAGREE (no line), single chime, MASTER CAUT. |
| 2 | Clear the failure. | The caution goes. |
| 3 | Activate **Engine 4 max thrust miscalculated (thrust loss)**; thrust levers 3 and 4 to FLX/MCT, hold 5 s. | Phase 3: amber ENG 4 THRUST LOSS with THR LEVER 4 ... IDLE and ENG 4 MASTER ... OFF (a red CONFIG PARK BRK ON also shows: normal with take-off power and the parking brake). |
| 4 | Levers back to IDLE, clear the failure. | The cautions go. |

## 18. E/WD N1 red above 111 % with a red cross

> **Aircraft:** A380X | **Where:** E/WD N1 indication | **Status:** Not sim-tested. Unit test N1.spec.ts.

### What it is for

FCOM DSC-70-90 N1 INDICATIONS: the N1 value turns red above the 111 % red limit, and a red cross stays next to it after an exceedance until the next engine start on the ground. The A380X E/WD N1 stayed green.

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **Hard to see in the sim.** | The A380X has no N1 overspeed failure; the N1 rarely passes 111 %. | - |

### How to use

1. Nothing to set. After an N1 exceedance the red cross stays until the next engine start on the ground.

### How to test

| # | Do this | Expected result |
|---|---|---|
| 1 | A380X, engines started on the ground, normal take-off thrust. | N1 values green, no red cross. |
| 2 | Run `npx vitest run fbw-a380x/src/systems/instruments/src/EWD/elements/N1.spec.ts`. | All tests pass (red above 111 %, the cross latched until the next ground start). |

## 19. Crossbleed engine start with X BLEED at AUTO

> **Aircraft:** A380X | **Where:** Overhead AIR panel (X BLEED selector, APU BLEED and ENG BLEED pbs); ENG MASTER levers; SD BLEED page | **Status:** Tested in the sim by the user on the integration build: OK, checked in the recorder log (2026-10-09 evening, FCOM card tests X1 and X2). Rust tests (pneumatic.rs, engine_out_aircraft_tests.rs). The core branch itself was not sim-tested.

### What it is for

A380 FCOM DSC-36-10-50 (I.94076-94079): the crossbleed valves "are normally closed. They open automatically, and at the same time: On ground, at engine start; When the APU bleed valve is selected open; In case any engine bleed system fails." The first case was missing: with the X BLEED selector at AUTO the valves stayed closed during an engine start, so the crossbleed engine start of PRO-SUP-70-10 (I.181024-181029: ALL ENG BLEEDS ON, X BLEED AUTO, SECOND ENG START) had no starter air without the APU. Now the three crossbleed valves open together on the ground while the FADEC commands the start valve of any engine open, and the running engines supply the starter air.

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **"At engine start" is from the ENG MASTER ON (or MAN START ON, crank) to the start valve closing at 58.4 % N3 or the end of the start.** | The FCOM gives no opening or closing point (design choice). | - |
| **On the ground only.** | FCOM rule. In flight a starter assisted relight without the APU still needs X BLEED OPEN (engine failures section). | Set X BLEED OPEN in flight. |
| **An engine bleed failure is modelled as an ENG BLEED pb OFF.** | There is no engine bleed failure with the pb ON yet. | - |
| **A full MSFS restart is needed after an update.** | The systems WASM module changes. | Restart the sim after installing. |

### How to use

1. Start the first engine with APU air as usual.
2. Set the APU BLEED pb OFF, keep all ENG BLEED pbs ON and the X BLEED selector at AUTO.
3. ENG START selector IGN START, then ENG MASTER ON for the next engine: the crossbleed valves open by themselves and close again after the start.

### How to test

From the FCOM card `reports/test-card-fcom-batch-2026-10-09.md`, "A380X crossbleed engine start". Setup: on the ground, PARKING BRK ON, APU running with the APU BLEED pb ON, engine 1 started and at idle, ENG MASTER 2, 3 and 4 OFF, ENG START selector IGN START, all four ENG BLEED pbs ON, X BLEED selector AUTO.

| # | Do this | Expected result |
|---|---|---|
| 1 | ECAM control panel: BLEED key. APU BLEED pb OFF; PACK 1 and PACK 2 pbs OFF; wait 10 s. | SD BLEED page: the three crossbleed valves closed. |
| 2 | ENG MASTER 2 ON; wait until engine 2 shows AVAIL (1 to 2 min). | Within 2 s the three crossbleed valves open; engine 2 N3 rises; no ENG 2 START FAULT. After the start the crossbleed valves are closed again. |

| | | |
|---|---|---|
| 3 | ENG MASTER 3 and ENG MASTER 4 ON together; wait until both are AVAIL. PACK 1 and 2 ON. | The crossbleed valves open within 2 s, engines 3 and 4 start without START FAULT, the valves close again. |

## 20. Engine alerts as in the FCOM

> **Aircraft:** A380X | **Where:** E/WD, STATUS, MASTER CAUT lights; SD ENGINE page | **Status:** Tested in the sim by the user on the integration build: OK (2026-10-09 evening, FCOM card tests A1 IGN / OIL FILTER levels and A2 START FAULT; 2026-10-10 morning, A3 EGT OVER LIMIT with FADEC FAULT). The in-flight START FAULT cases and the deferred procedures: not sim-tested (unit tests EngineFailAlerts, EngineStartAlerts, EngineParameterAlerts, EngineOilAlerts, EngineFuelFilterAlerts specs; Rust tests). The core branch itself was not sim-tested.

### What it is for

The engine alerts of this branch are aligned with A380 FCOM PRO-ABN-ECAM-10-70 and its figures (levels, inhibitions, SD pages, lines):

- **ENG n FAIL** is no longer inhibited from V1 to 1500 ft (the FCOM figure has no inhibition). Its line AUTO RELIGHT IN PROGRESS shows only while the FADEC auto relight runs.
- The **IGNITION** memo also shows for the continuous ignition the FADEC selects by itself (auto relight, quick relight).
- **ENG n START FAULT**, **IGN A (B) FAULT**, **OIL FILTER CLOGGED** and **FUEL FILTER CLOGGED** come without chime and without MASTER CAUT (crew awareness). IGN A+B FAULT keeps the single chime and MASTER CAUT.
- **ENG n EGT OVER LIMIT** and **N1/N2 OVER LIMIT** call no SD page. EGT OVER LIMIT does not come up with an ENG n FADEC FAULT (FCOM ENG FADEC FAULT: the EGT OVER LIMIT alert will not trigger in that case).
- In flight, **START FAULT NO STARTER AIR PRESSURE** comes up with its windmill start lines (WINDMILL START INITIATED, ENG PARAMETERS MONITOR, IF ABNORMAL: ENG n MAN START OFF, ENG n MASTER OFF), and **NO LIGHT UP** comes after the 30 s of the relight (ENG RELIGHT IN FLIGHT), not after the 20 s of the ground start. The shared engine start schedule got two fields for this; the A32NX keeps its values.
- New deferred procedures: **ENG TWO ENGS OUT** (ALL PHASES: IF GW DECREASE RQRD, JETTISON ARM ON, JETTISON ACTIVE ON), **TWO ENGS OUT ON SAME SIDE** (FOR APPROACH) and **ON OPPOSITE SIDE** (APPR & LDG), shown under their heading on the STATUS page. **ENG n START VLV FAULT (NOT CLOSED)** gets its LIMITATIONS, INOP SYS and, for engine 1 or 4, the FOR APPROACH deferred procedure IF SEVERE ICE ACCRETION: FOR LDG : FLAP LVR 3 (l.173136-173158).

> **NOTE**
> On this branch the JETTISON ARM / ACTIVE lines of the TWO ENGS OUT deferred procedures read the FUEL JETTISON pbs directly; with `core/fuel` merged they read its jettison FWS states.

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **A full MSFS restart is needed after an update.** | The systems WASM module changes (engine start schedule). | Restart the sim after installing. |

### How to use

1. Nothing to set: follow the E/WD as usual. An alert without chime or MASTER CAUT is a crew awareness alert.
2. With two engines out, check the deferred procedures on the STATUS page before the approach.

### How to test

From the FCOM card `reports/test-card-fcom-batch-2026-10-09.md`, "A380X engine alert corrections" T1 to T3.

| # | Do this | Expected result |
|---|---|---|
| 1 | On the ground, PARKING BRK ON, four engines at idle, no alert. flyPad > Failures > ATA 74 > "Engine 1 igniter A": activate, wait 10 s. | Amber ENG 1 IGN A FAULT, no single chime, both MASTER CAUT lights off. |
| 2 | ATA 79 > "Engine 1 oil filter clog": activate, wait 10 s. | Amber ENG 1 OIL FILTER CLOGGED, no chime, no MASTER CAUT. |
| 3 | ATA 74 > "Engine 1 igniter B": activate. | Amber ENG 1 IGN A+B FAULT with a single chime and both MASTER CAUT lights. Press a MASTER CAUT pb and deactivate the three failures. |
| 4 | APU BLEED ON, engine 1 running, ENG MASTER 2 OFF, ENG START IGN START. ATA 80 > "Engine 2 hot start (start EGT over limit)": activate; ENG MASTER 2 ON; wait about 60 s. | Amber ENG 2 START FAULT with EGT OVERLIMIT, no chime, MASTER CAUT off. ENG MASTER 2 OFF, deactivate the failure. |
| 5 | In flight above FL100, A/THR ON, levers in CL. ATA 73 > "Engine 1 FADEC network link": activate; press a MASTER CAUT pb. ATA 72 > "Engine 1 EGT overtemperature": activate; wait 60 s. | ENG 1 FADEC FAULT with chime and MASTER CAUT; then the engine 1 EGT rises and turns amber, but no ENG 1 EGT OVER LIMIT (no new chime). Deactivate both. |

## 21. Braking and steering control failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 32 Landing Gear; brake pedals, tiller, PARK BRK; E/WD, STATUS; triple brake pressure indicator | **Status:** Recorded as passed in the commit message (batch 4 test B1, integration build, 2026-10-09/10 sessions); the session notes have no separate entry for it. Unit tests FwsBrakeAlerts.spec.ts, a380.spec.ts; Rust tests. The core branch itself was not sim-tested.

### What it is for

Eight new failures of the braking control system (BCS), the emergency brake control unit (EBCU) and the steering control system (SCS), with the BRAKES and STEER alerts of A380 FCOM PRO-ABN-ECAM-10-32. The Rust hydraulics choose the braking mode the faults leave (DSC-32-10-30) and the nose wheel steering supply (DSC-32-10-40).

| flyPad failure (ATA 32) | What it does | ECAM |
|---|---|---|
| Brakes normal braking (BCS NORM BRK) | The pedals brake in alternate braking (ALTN circuit, A-SKID available). | BRAKES NORM BRK FAULT |
| Brakes alternate braking (BCS ALTN BRK) | Alone: no effect while normal braking works. With the normal braking failed: emergency braking through the EBCU (no A-SKID, pressure limited). | BRAKES ALTN BRK FAULT; with the EBCU failed too: ALTN + EMER BRK FAULT |
| Brakes emergency braking (EBCU) | The EBCU is lost; with normal and alternate braking failed too, no pedal braking: the parking brake is the only braking. | BRAKES EMER BRK FAULT; all three: BRAKES PEDAL BRAKING FAULT (replaces the single alerts) |
| Steering normal nose wheel steering (NORM N/W STEER) | The nose wheel steers on the alternate system (steering LEHGS), at most 10 deg/s instead of 15 deg/s. | STEER NORM N/W STEER FAULT |
| Steering alternate nose wheel steering (steering LEHGS) | No alternate NWS: with the normal NWS also lost, the nose wheel casters. | STEER ALTN N/W STEER FAULT |
| Steering body wheel steering (B/W STEER) | The body wheel steering stays centred and locked. | STEER B/W STEER FAULT (crew awareness) |

| | | |
|---|---|---|
| Steering control system 1 (STEER CTL 1) / 2 (STEER CTL 2) | One SCS is enough (the backup takes over); with both lost neither NWS nor BWS works. | STEER CTL 1 (2) FAULT; both: STEER N/W + B/W STEER FAULT |

### Also in this feature

- **BRAKES ACCU PRESS LO** (accumulator below 3 000 psi for 10 s, design value) and **PARK BRK PRESS LO** (park brake on and pressure below 1 700 psi for 5 s, design value).
- The second B/W STEER FAULT text is renamed **B/W STEER NOT LOCKED**, as in the FCOM.
- With only engines 3 and 4 running (no GREEN), the nose wheel now steers on the alternate system instead of not at all, as the FCOM asks.

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **A brake fault acts on all wheel groups at once.** | The model has one NORM circuit (GREEN) and one ALTN circuit (YELLOW, one accumulator) for all wheel groups; the real A380 splits them per gear. | - |
| **The nose wheel steering request keeps the A-SKID condition of this base.** | The FCOM batch 5 change that drops it is on `core/fcom-audit-fixes`. | Merge that branch for it. |
| **A full MSFS restart is needed after an update.** | The systems WASM module changes. | Restart the sim after installing. |

### How to use

1. Open flyPad > Failures > ATA 32 Landing Gear and activate a brake or steering failure on the ground.
2. Follow the E/WD; brake with the pedals (watch the triple brake pressure indicator) and steer with the tiller.
3. Deactivate the failure.

### How to test

From batch 4 test B1 (`reports/test-card-batch4-2026-10-10.md`). Setup: on the ground, four engines running, park brake OFF, below 10 kt.

| # | Do this | Expected result |
|---|---|---|
| 1 | Activate "Steering normal nose wheel steering (NORM N/W STEER)"; turn the tiller fully left. | Amber STEER NORM N/W STEER FAULT with a chime; the nose wheel turns visibly slower. |
| 2 | Deactivate it; stop. Activate "Brakes normal braking (BCS NORM BRK)"; press both brake pedals fully. | BRAKES NORM BRK FAULT (no chime); triple indicator L/R about 2 500 psi. |
| 3 | Also activate "Brakes alternate braking (BCS ALTN BRK)"; press both pedals. | BRAKES ALTN BRK FAULT too; L/R about 1 160 psi (emergency braking). |
| 4 | Deactivate both; PARK BRK ON. | The alerts go; normal braking. |

## 22. Erroneous ADR airspeed and AOA failures, PRIM air data monitoring

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 34 Navigation; PFD; E/WD, STATUS | **Status:** Tested in the sim by the user on the integration build: OK (2026-10-09/10 night flight, batch 4 test A1: one, then two ADRs wrong; the AP could not be re-engaged afterwards, as the FCOM says: the FD needs two ADRs). The AOA failures: not sim-tested. Unit tests FwsAirDataAlerts.spec.ts, FwsLimitations.spec.ts, native air_data_rejection_test, Rust tests. The core branch itself was not sim-tested.

## What it is for

Six new failures: **ADR 1 (2)(3) airspeed erroneous (pitot)** and **ADR 1 (2)(3) angle of attack erroneous**. The ADR then sends a wrong value with a valid status (airspeed 40 kt low, AOA 10 deg high; design values above the FCOM monitoring thresholds), which the PRIM model could never reject, as it only votes on the status.

A new systems module compares the three ADRs with the standby air data as the PRIMs do (A380 FCOM DSC-34-10-30-10 / -40) and rejects a diverging source until the end of the flight, or all of them when it cannot tell which one is wrong. A rejected ADR goes to the PRIM and SEC models as failed: alternate law with two ADRs out, direct law with all of them.

| Alert (PRO-ABN-ECAM-10-34-10) | When | Lines |
|---|---|---|
| NAV AIR DATA DISAGREE | One or more sources rejected (not all). | ADR n (or STBY INSTRUMENTS) REJECTED BY PRIMs, RISK OF UNDUE OVERSPEED WARNING (two sources), AIR DATA XCHECK, AIR DATA SWTG AS RQRD. |
| NAV ALL AIR DATA DISAGREE | All air data rejected. | ALL AIR DATA REJECTED BY PRIMs, the thrust and pitch target lines of the FCOM tables for the altitude, FLAPS & L/G MAINTAIN CURRENT CONF, ... FLY THE GREEN, FLY PITCH AND THRUST. |
| NAV AOA DISAGREE | Two or more AOAs rejected (one alone gives no NAV alert). | RISK OF UNDUE STALL WARNING. |

Also: F/CTL DIRECT LAW MAX SPEED 310/.86 and MANEUVER WITH CARE are now APPR & LDG limitations, as in the FCOM (PRO-ABN-ECAM-10-27-10); a limitation shown in ALL PHASES by another alert is not repeated in APPR & LDG.

## Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **A rejected ADR stays rejected until the end of the flight, also after the failure is cleared.** | As the PRIMs do (FCOM). | A new flight. |
| **The cabin altitude table, the MFP heating lines and the level-off thrust table above FL 250 of ALL AIR DATA DISAGREE are not shown.** | Not modelled. | - |
| **A full MSFS restart is needed after an update.** | The systems and fbw WASM modules change. | Restart the sim after installing. |

## How to use

1. Open flyPad > Failures > ATA 34 Navigation and activate an ADR airspeed or AOA failure in flight.
2. Follow the E/WD (cross-check the air data, switch the AIR DATA source as required).
3. Deactivate the failure; the PRIMs keep the ADR rejected for the rest of the flight.

## How to test

From batch 4 test A1 (`reports/test-card-batch4-2026-10-10.md`). Setup: cruise above FL250, AP ON.

| # | Do this | Expected result |
|---|---|---|

| | | |
|---|---|---|
| 1 | Activate "ADR 1 airspeed erroneous (pitot)"; wait 5 s. | The CAPT PFD speed reads 40 kt low; amber NAV AIR DATA DISAGREE with ADR 1 REJECTED BY PRIMs; normal law stays. |
| 2 | Activate "ADR 2 airspeed erroneous (pitot)"; wait 5 s. | F/CTL ALTN LAW (PROT LOST); the AP disconnects. |
| 3 | Deactivate both. | The ADRs stay rejected until the end of the flight (normal). |
| 4 | New flight, cruise. Activate "ADR 1 angle of attack erroneous" and "ADR 2 angle of attack erroneous". | Amber NAV AOA DISAGREE with RISK OF UNDUE STALL WARNING. |

## 23. FCOM batch 7: RCL / RCL LAST, T.O CONFIG NORMAL, automatic SD pages, BTV INOP

> **Aircraft:** A380X | **Where:** ECAM control panel (RCL, RCL LAST, T.O CONFIG); E/WD; SD; auto brake | **Status:** Recorded as passed in the commit message on the integration build (batch 2 tests N3 T.O CONFIG NORMAL, N4 RCL / RCL LAST, N6 automatic SD pages, 2026-10-09/10 sessions; the session notes give N6 CRUISE at the first take-off). BTV INOP: not sim-tested. Unit tests FwsRecall.spec.ts, ToConfigNormal.spec.ts, FwsSystemDisplayLogic.spec.ts; Rust autobrake tests. The core branch itself was not sim-tested.

### What it is for

- **RCL LAST** brings back the alert of the last CLR action (both pbs wrote the same variable before). **RCL** brings back every cleared alert at once, or NORMAL for 3 s when there is none; only a press of more than 3 s shows the alerts inhibited by the flight phase, so a short press no longer removes the T.O INHIBIT memo for the flight (A380 FCOM DSC-31-40-20).
- **T.O CONFIG NORMAL** follows the A380 checks of DSC-31-40-10 (the FCOM list of T.O CONFIG alerts: surfaces and trims, doors, BRAKES HOT, the generators and drives of the four engines, engine oil temperature, PITCH TRIM/FMS/CG and T.O FLAPS/FMS DISAGREE, T.O speeds) instead of the A320 GEN 1+2 / two doors condition.
- **Automatic SD pages** (DSC-31-40-30): CRUISE from 1500 ft after take-off (it came 60 s after the flaps were up), WHEEL at the landing gear extension at any altitude (only below 16000 ft before).
- **BTV INOP**: while an ECAM alert lists BTV in STATUS INOP SYS (ENG 2 (3) SHUT DOWN, TWO ENGS OUT, ENG 2 (3) REVERSER faults), BTV does not arm, and an armed BTV reverts to BRK HI (DSC-32-20-220, LIM-32).

### Limitations and why

| Limitation | Why | What to do |
|---|---|---|
| **T.O CONFIG NORMAL does not check the ELEC GEN / DRIVE alerts on this branch.** | Those alerts are on `core/a380x-elec`. | Use `core/a380x-elec-fuel-alerts` (both branches merged). |
| **CONFIG L (R) SIDESTICK FAULT (BY TAKE-OVER) is not checked.** | The sim has no sidestick take-over fault. | - |
| **A full MSFS restart is needed after an update.** | The systems WASM module (autobrakes) and the ECAM control panel XML change. | Restart the sim after installing. |

### How to use

1. RCL LAST: bring back the last cleared alert. RCL: bring back all cleared alerts; hold it more than 3 s to see the alerts inhibited by the flight phase.
2. Before take-off press T.O CONFIG: T.O CONFIG NORMAL shows when no alert of the FCOM list is active.
3. The SD changes page by itself; with an alert that lists BTV INOP, select another auto brake mode.

## How to test

Adapted from batch 2 tests N3, N4 and N6 (`reports/test-card-batch2-2026-10-10.md`): the card used the generator failures, whose ELEC alerts are not on this branch; the feed pump failures of this branch give two cautions instead.

| # | Do this | Expected result |
|---|---|---|
| 1 | On the ground, engines running and warm, T.O speeds entered, pitch trim in range, doors closed, flaps 0. ECAM control panel: T.O CONFIG key. | A CONFIG alert for the flaps; no T.O CONFIG NORMAL. |
| 2 | Set the take-off flaps of the FMS; T.O CONFIG key again. | T.O CONFIG NORMAL. |
| 3 | flyPad > Failures > ATA 28: "Feed tank 1 main pump", then "Feed tank 2 main pump". CLR until both cautions are cleared. Press RCL LAST. | Only the last cleared caution comes back. |
| 4 | CLR. Press RCL (short). | Both cautions come back at once. |
| 5 | CLR both, deactivate both failures, wait 5 s, press RCL. | NORMAL for 3 s. |
| 6 | Take off with the SD on its automatic page. | The T.O INHIBIT memo still shows at take-off after the RCL presses; the SD shows CRUISE as soon as passing 1500 ft. |
| 7 | On the approach, extend the landing gear. | The SD shows WHEEL at once, also above 16000 ft. |
| 8 | Optional, on landing with BTV armed: set ENG MASTER 2 OFF in flight before. | STATUS INOP SYS BTV; BTV does not arm (an armed BTV reverts to BRK HI). |